MILITARY SPECIFICATION

TRANSMITTING SET, RADIO, AN/FRT-84B(V)

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the design, construction, and performance requirements for radio transmitter AN/FRT-84B(V), hereinafter referred to as the transmitter. The transmitter is for general-purpose use ashore on Navy and Defense Communications System (DCS) high-frequency (HF) long-haul independent sideband (ISB) and single sideband (SSB) radio communication circuits. The transmitter is a 10 kilowatt (kW) peak envelope power (PEP), frequency-synthesized, forced air-cooled radio transmitter that features complete automatic tuning to the frequency selected at the MD-777A/FRT Modulator-Synthesizer. The transmitter is capable of remote control when equipped with the AN/FSQ-98(V) or the AN/FSQ-98A(V) Remote Control System (RCS). Modular construction permits removal and replacement of modules, assemblies, and subassemblies by maintenance personnel. The transmitter operating frequency range is from 2.0000 megahertz (MHz) to 29.9999 MHz in 100 hertz (Hz) steps.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-365	Silver Plating, Electrodeposited; General Requirements For

MILITARY

MIL-T-27	Transformer And Inductor (Audio, Power, and High Power Pulse), General Specification For
MIL-P-116	Preservation, Methods Of
MIL-E-15090	Enamel, Equipment, Light-Gray (Formula No. 11)
MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship and Shore: General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Repair Parts, Packaging And Packing Of
MIL-F-19207/24	Fuseholders, Extractor Post Type, Blown Fuse Indicating, Type FHL35W
MIL-S-19500/420	Semiconductor Device, Diode, Silicon, Power Rectifier, General Purpose Tx And Non-Tx Types 1N5550 Through 1N5554
MIL-S-19500/427	Semiconductor Device, Diode, Silicon, Power Rectifier Types 1N5614, 1N5616, 1N5618, 1N5620, 1N5622, Tx And Txv
MIL-E-21981	Electronics Type Designations, Identification Plates And Markings, Requirements For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (ELEX-8111), Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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SPECIFICATIONS (Cont)

MILITARY

MIL-T-28784	Transmitter, Radio, AN/FRT-85(V) 40 kW Pep High Frequency, High Stability Independent Sideband
MIL-O-39021	Ovens, Crystal

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-189	Rack, Electrical Equipment, 19-Inch And Associated Panels
MIL-STD-415	Test Provisions For Electronic Systems And Associated Equipment, Design Criteria For
MIL-STD-449	Radio Frequency Spectrum Characteristics, Measurement Of
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-471	Maintainability Verification/Demonstration/Evaluation
MIL-STD-781	Reliability Design Qualification And Production Acceptance Tests: Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-965	Parts Control Program
MIL-STD-1353	Electrical Connectors, Plug-In Sockets And Associated Hardware, Selection And Use Of
MIL-STD-1364	Standard General Purpose Electronic Test Equipment
MIL-STD-1378	Requirements For Employing Standard Electronic Modules
MIL-STD-1472	Human Engineering Design Criteria For Military Systems, Equipment And Facilities

HANDBOOK

MIL-HDBK-217	Reliability Prediction Of Electronic Equipment

(Copies of specifications, standards, and handbooks required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

S1.4-1961	American National Standards For Sound Level Meters

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

Standard RS 225-59	Rigid Coaxial Transmission Lines 50 Ohm

(Application for copies should be addressed to the Electronic Industries Association, 2001 Eye Street, N.W., Washington, DC 20006.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70-1978	National Electrical Code

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(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 General. The transmitter shall be in accordance with MIL-E-16400 to the extent specified herein.

3.1.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Description. The transmitter shall consist of an Interconnecting Box, J-2822B/FRT-84(V); an Amplifier, Radio Frequency, AM-6046B/FRT; a Power Supply, PP-6067A/FRT; an Amplifier, Radio Frequency, AM-6047B/FRT-84(V) with associated power supply; a Modulator-Synthesizer, MD-777A/FRT; and a Keyer, Frequency Shift, KY-655A/FRT, and associated cabinets. All of these major components shall be contained in two equipment cabinets that can be installed to give the appearance of a single unit. The transmitter shall be remotely controllable when equipped with the AN/FSQ-98(V) or an AN/FSQ-98A(V) RCS. The MD-777A/FRT, hereinafter referred to as the exciter, shall determine the output frequency stability when operated on the internal frequency standard and furnish to the AM-6046B/FRT an output signal of any level from a nominal minimum of 10 milliwatts (mW) to a maximum of 250 mW when terminated into a nominal 50-ohm impedance. The KY-655A/FRT shall provide continuous wave (CW), single channel frequency shift keying (FSK) and facsimile (FAX) keying to the exciter upper sideband (USB), channel A_1, audio input. The AM-6046B/FRT shall be a two-stage, nominal 40 decibel (dB) linear power amplifier capable of producing an output of 1 kW PEP with a nominal input of 100 mW. The AM-6047B/FRT-84(V) shall be a linear power amplifier capable of producing an output of 5 kW average (avg) and 10 kW PEP with a maximum nominal input of 800 watts (W) PEP (two-tone). The interconnecting box shall contain system control circuitry for sequencing and coordination of the operation of transmitter units and shall provide a central location for interfacing of the various transmitter units. Automatic control circuits, in conjunction with the exciter and the AM-6047B/FRT-84(V), shall compensate for variations in system gain or loading to optimize performance and protect against overload and the generation of excessive distortion. The PP-6067A/FRT shall produce high voltage (B-plus) for the AM-6046B/FRT from 220 volts alternating current (VAC), three-phase, 47 Hz to 63 Hz, power from the autotransformer in the bottom of the cabinet containing the AM-6046B/FRT. The power supply associated with the AM-6047B/FRT-84(V) shall produce operating voltages for the AM-6047B/FRT-84(V) 10 kW amplifier from 220 VAC, three-phase, 47 Hz to 63 Hz, power from the autotransformer.

3.2.1 Parts, materials, and processes. Parts, materials, and processes shall be in accordance with MIL-E-16400 to the extent specified herein.

3.2.1.1 Parts control. The parts to be incorporated in the transmitter shall be controlled in accordance with MIL-STD-965, Procedure I, except that those parts listed under the Military Parts Control Advisory Group (MPCAG) Functions, Mechanical Parts are exempt.

3.2.1.2 Individual selection of parts. Transmitter performance shall not be dependent on the selection of individual parts unless such selection is specifically approved by the procuring activity. If such individual part selection is approved, sufficient information and data shall be included in applicable technical publications and engineering drawings to allow designated repair activities to perform such selection.

3.2.1.3 Nonstandard parts. Only standard parts or approved nonstandard parts shall be employed in the transmitter. Nonstandard parts shall be approved in accordance with MIL-STD-965, Procedure I, prior to the submission of the first article sample (FAS) for Government inspection and approval. The FAS shall incorporate only standard parts and approved nonstandard parts prior to conducting the first article inspection.

3.2.1.3.1 Previous approval of nonstandard parts. Nonstandard parts previously granted approval for the AN/FRT-84B(V) and the AN/FRT-83B(V) shall be considered to be standard parts for the purposes of this specification.

3.2.1.3.2 Quality inspection of approved nonstandard parts. All approved nonstandard parts shall be subject to quality conformance inspection, Group A only, in accordance with their pertinent component specification as specified in MIL-E-16400.

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3.3 Design.

3.3.1 General design requirements. The transmitter shall be assembled in two cabinets that can be installed side by side to give the appearance of a single unit. The cabinet containing the AM-6046B/FRT and the exciter, hereinafter referred to as the exciter-driver cabinet, shall also contain the J-2822B/FRT-84(V), the PP-6067A/FRT, the KY-655A/FRT, and, when the transmitter is so equipped, the local control unit (LCU) of the RCS. The other cabinet, hereinafter referred to as the final power amplifier cabinet, shall contain the AM-6047B/FRT-84(V) and its associated power supply.

3.3.1.1 Exciter-driver cabinet equipment arrangement. The exciter-driver cabinet and the equipment it contains shall be arranged in the following sequence from the top to the bottom of the cabinet:

a. Air intake and alarm panel

b. Blank panel

c. Exciter

d. KY-655A/FRT

e. LCU (when so equipped, or blank panel)

f. J-2822B/FRT-84(V)

g. AM-6046B/FRT

h. PP-6067A/FRT

i. Protective screen

j. Circuit breaker panel

3.3.1.2 Primary power. The primary power requirement for the transmitter shall be less than 30 kilovolt-amperes (kVA) (27 kW). The power factor shall be 0.90 or greater when the transmitter is operated in any operating mode specified herein at any carrier frequency within the specified frequency range.

3.3.1.2.1 Primary power cable entry. Primary power cable shall enter the transmitter through the bottom of the exciter-driver cabinet.

3.3.1.2.2 Primary power connections. The three-phase primary power from the local power source shall be connected through a box terminal lug type terminal board to a circuit breaker behind the exciter-driver cabinet circuit breaker panel using size 6 American Wire Gage (AWG) wire. Single-phase primary power from the local power source shall be connected through a terminal board to a circuit breaker behind the exciter-driver cabinet circuit breaker panel.

3.3.1.2.3 Exciter-driver cabinet primary power circuit breaker. The exciter-driver cabinet primary power circuit breaker shall function primarily as a readily available emergency shutoff.

3.3.1.2.4 Autotransformer. An autotransformer located in the bottom of the exciter-driver cabinet shall furnish primary power for all transmitter units except the exciter, KY-655A/FRT, and, when installed, the LCU.

3.3.1.2.4.1 Autotransformer input. Power from the exciter-driver cabinet circuit breaker shall connect to the autotransformer input through a terminal board with labeled taps that shall provide for transmitter operation from any single alternating current (AC) power source of 200, 208, 220, 230, 240, 440, 460, or 480 VAC, each ±10 percent, 47 Hz to 63 Hz, three-phase. The input to the autotransformer shall be configured as a wye with a floating neutral.

3.3.1.2.5 AM-6047B/FRT-84(V) primary power. Primary power for the AM-6047B/FRT-84(V) and its associated power supply shall be 220 VAC, three-phase, 47 Hz to 63 Hz, from the autotransformer through the terminal board specified in 3.3.1.2.4.1 and an intervening terminal board to a 90-ampere (A) circuit breaker behind the final power amplifier cabinet circuit breaker panel. The three-phase primary power input circuit to the AM-6047B/FRT-84(V) plate-screen high-voltage transformer shall be delta configured. The AM-6047B/FRT-84(V) plate-screen transformer output

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wires shall be strain-relieved by mechanical supports at the exit point, as in the Government-furnished equipment (GFE) (see 6.5). Filament supply wiring shall be size 2 AWG. Additional output terminals on the autotransformer shall furnish three-phase 20.6 VAC and single-phase 115 VAC to the final power amplifier cabinet and single-phase 115 VAC to the J-2822B/FRT-84(V) through an intervening terminal board. The 115-VAC circuit to the J-2822B/FRT-84(V) shall be protected by a 7.5 A circuit breaker behind the exciter-driver cabinet circuit breaker panel.

3.3.1.2.6 PP-6067A/FRT primary power. Primary power for the PP-6067A/FRT, the power supply for the AM-6046B/FRT, shall be 220 VAC, three-phase, 47 Hz to 63 Hz, from the auto-transformer through the terminal board specified in 3.3.1.2.4.1. The three-phase primary power input circuit to the plate transformer of the PP-6067A/FRT shall be configured as a wye with a floating neutral.

3.3.1.2.7 Exciter, KY-655A/FRT and LCU primary power. Primary power for the exciter, KY-655A/FRT, and the LCU, when installed, shall be provided by a separate local power source of 115 VAC or 230 VAC, single-phase, 47 Hz to 63 Hz.

3.3.1.2.7.1 Exciter, KY-655A/FRT and LCU primary power distribution. AC power to the exciter, KY-655A/FRT, and the LCU shall be distributed by means of a standard multiple outlet strip mounted inside the rear of the exciter-driver cabinet using grounding type outlets.

3.3.1.2.8 Circuit breakers. The three-phase primary power inputs to the exciter-driver cabinet and the final power amplifier cabinet shall each be connected through a 90 A circuit breaker mounted on the rear of the applicable cabinet circuit breaker panel. The internal single-phase 115 VAC primary power circuit shall be connected through 7.5 A circuit breakers mounted on the rear of the exciter-driver cabinet and the final power amplifier cabinet breaker panels. The external single-phase 115 VAC primary power circuit shall be connected through a 3.5 A circuit breaker mounted on the rear of the exciter-driver cabinet circuit breaker panel. The circuit breaker and associated terminal strips on the line side of the circuit breakers shall be enclosed to prevent accidental contact by personnel when working in the cabinet. Circuit breakers shall serve as the main power switches except the transmitter shall normally be turned ON and OFF by the amplifier OFF control specified in 3.3.2.19.5. Circuit breakers shall provide a visible indication when tripped.

3.3.1.3 Accessibility. The transmitter shall be designed so that parts, terminals, and wiring are accessible for circuit-checking and adjustment, maintenance, repair, and replacement with a minimum number of special tools (see 6.6). Parts shall be mounted so that identification markings are readily visible with minimum disassembly of the transmitter.

3.3.1.4 Use of electron tubes and solid-state devices. Solid-state devices shall be employed instead of electron tubes to the maximum extent consistent with the performance requirements of this specification and for general specification application. The exciter shall use only solid-state devices. The number of different types of solid-state devices and electron tubes shall be kept to a minimum. All electron tubes and solid-state devices shall be operated well below their maximum specified rating to ensure conformance to the mean-time-between-failure (MTBF) requirement specified in 3.4.1. Where applicable, overload protection shall be provided. MIL-E-16400 shall be used as a guide in the selection of solid-state devices.

3.3.1.5 Capacitors.

3.3.1.5.1 High-voltage feedthrough capacitors. High-voltage feedthrough capacitors which may collect or be subjected to condensed moisture shall be fitted with insulating shrouds to prevent arc-over.

3.3.1.5.2 Neutralizing capacitors. Neutralizing capacitors shall be vacuum or gas-filled ceramic type and shall be rated to withstand at least two times the maximum voltage and current that might be encountered under worst-case operating conditions.

3.3.1.5.3 Feedback capacitors. Variable capacitors used for negative feedback to improve linearity shall be locked at the correct value during manufacture. However, provision for adjustment shall be included to permit compensation for circuit component aging and replacement.

3.3.1.5.4 Variable capacitors. Servo-controlled variable capacitors used for the higher power tuning elements shall be rated to withstand at least 1.5 times the maximum voltage and current encountered under the worst-case operating conditions specified herein when the transmitter is operating on any frequency between the limits of 2.0000 MHz to 29.9999 MHz.

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3.3.1.6 Type BNC and Type N radio frequency (RF) coaxial connectors. Where type BNC or N RF coaxial connectors are specified herein, connectors shall be selected in accordance with MIL-STD-1353.

3.3.1.7 Transformers. All provisions of MIL-T-27 shall apply for all transformers employed in the design of this transmitter.

3.3.1.8 PP-6067A/FRT power supply and AM-6047B/FRT-84(V) screen rectifiers. Type JANTX 1N5554, 1000 volts (V), 3 A, 100 A surge, high-voltage rectifiers in accordance with MIL-S-19500/420 shall be used in the PP-6067A/FRT 2400 V and 1000 V power supplies and in the screen circuit of the AM-6047B/FRT-84(V).

3.3.1.9 Insulating compounds. Compounds which soften and flow at any temperature less than +75° Celsius (C) shall not be used unless contained in such a manner that the compound will not flow from the container during service use. Compounds which crack at any temperature above -62°C shall not be used.

3.3.1.10 Resistors. Fixed composition resistors shall be of ±10 percent or better tolerance. Resistors having a tolerance of less than ±5 percent shall be limited to cases where circuit use makes them mandatory to conform to the performance requirements of this specification.

3.3.1.11 Encapsulation. Plastic encapsulated transistors or microelectronic devices shall not be used unless specifically approved, in writing, by the procuring activity.

3.3.1.12 Semiconductors. All semiconductors used in the transmitter shall be silicon.

3.3.1.13 Cables. Cables supplied with the equipment shall be fabricated in accordance with Requirement 66 of MIL-STD-454.

3.3.1.14 Unacceptable materials. The transmitter shall conform to the Unacceptable materials paragraph of MIL-E-16400. Aluminum wire shall not be used.

3.3.1.15 Electrical contacts. Friction or pressure contacts, when used, shall be designed to prevent erratic operation caused by the effects of vibration and temperature. Whenever practicable, electrical switch and relay contacts shall be of the self-cleaning type.

3.3.1.16 Relays. Relays which are subject to accelerated deleterious effects caused by humidity shall be hermetically or vacuum sealed. This requirement shall not apply to relays K-1 and K-2 in the band relay assembly (9A1A16) or to relay K-1, standby-operate relay, in assembly 1A1.

3.3.1.17 Thermal design. The thermal design of the transmitter shall be in accordance with the requirements of the Thermal design paragraph (and all subparagraphs) of MIL-E-16400, except Thermal instrumentation. The cooling air inlet-to-outlet temperature differential shall not exceed 60°C.

3.3.1.17.1 Ventilation and cooling. The transmitter shall be ventilated and cooled in such a manner that under normal conditions of operation at the maximum ambient temperature of 50°C no part shall reach a temperature which will cause damage. Forced air-cooling with replaceable air filters provided at each air inlet shall be used. Filters shall be removable and replaceable while the transmitter is in operation without affecting normal operation or endangering operating personnel. The cooling system shall limit the maximum junction temperature of solid-state components and the seal temperatures of the AM-6046B/FRT and AM-6047B/FRT-84(V) tubes to temperatures below that recommended by the component manufacturer under the environmental conditions specified in 3.8. The forced air system shall be capable of overcoming the added back pressure of 15 meters (m) (approximately 49.1 feet (ft)) of metallic ducting to an exhaust port and 3 m (10 ft) of ducting to an intake port.

3.3.1.17.1.1 Exciter-driver cabinet fans. Those components of the exciter-driver cabinet that require cooling shall be cooled by a cabinet flushing fan and an AM-6046B/FRT cooling fan.

3.3.1.17.1.1.1 Exciter-driver cabinet flushing fan. The exciter-driver cabinet flushing fan shall operate from a single-phase 115-VAC power source and shall be capable of delivering at least 400 cubic feet per minute (CFM) into a pressure head equivalent to 2.54 centimeters (cm) (1 inch (in.)) of water. Inlet air shall enter the cabinet through a filter in the rear

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cabinet door. Exhaust air shall exit out the top of the rear cabinet door. The cabinet flushing fan shall provide flushing air for the PP-6067A/FRT and forced air cooling for the exciter.

3.3.1.17.1.1.1.1 Cabinet flushing fan alarm. The cabinet flushing fan shall be equipped with an alarm that is activated by an airflow switch in the fan shroud when the cabinet flushing fan cooling system fails. Alarm indicators consisting of a visual indicator and an aural indicator shall be located on the air intake and alarm panel of the exciter-driver cabinet.

3.3.1.17.1.1.2 AM-6046B/FRT cooling fan. The AM-6046B/FRT cooling fan shall operate on 400 Hz alternating current (AC) converted by a self-contained inverter and shall be capable of delivering at least 120 CFM as installed in the GFE. Inlet air shall enter the front panel of the AM-6046B/FRT through a filter. The front door of the exciter-driver cabinet shall be equipped with acoustic baffling for the inlet air. Exhaust air shall exit through the top of the cabinet. A bolt hole pattern for mounting exhaust air ducting shall be provided in the top of the cabinet as in the GFE.

3.3.1.17.1.2 Final power amplifier cabinet blower. The final power amplifier cabinet blower shall operate from three-phase, 220 VAC, 47 Hz to 63 Hz power and shall be capable of delivering at least 1000 CFM at sea level. Inlet air shall enter the rear of the cabinet through a filter. Ducting of inlet and exhaust air shall conform to 3.3.1.17.1.3.

3.3.1.17.1.3 Ducting. Ducting shall have the same or larger cross-sectional area as the mating openings of the transmitter and shall have a smooth interior surface which minimizes resistance to air flow without special conditioning or treatment.

3.3.1.17.1.3.1 Final power amplifier cabinet ducting. The final power amplifier cabinet shall be fitted for ducting of both the intake and exhaust air at a rate of not less than 1000 CFM at sea level.

3.3.1.17.1.3.1.1 Intake duct. The intake duct shall be nominally 66 cm by 66 cm (26 in. by 26 in.) and shall be located on the rear of the final power amplifier cabinet as in the GFE. The duct flange shall project 2.8 cm (1.10 in.) beyond the rear of the cabinet to provide for connecting to the external ducting.

3.3.1.17.1.3.1.2 Exhaust duct. The exhaust duct shall be nominally 24 cm by 24 cm (9.5 in. by 9.5 in.) and shall be located on the top of the cabinet as in the GFE. The exhaust air outlet shall be provided with a bolt hole pattern as in the GFE for connecting to external ducting.

3.3.1.17.1.3.2 Exciter-driver cabinet ducting. The exciter-driver cabinet shall be fitted for ducting only the exhaust air at a rate of not less than 120 CFM as installed in the GFE. The exhaust duct shall be nominally 10.8 cm by 28.6 cm (4.25 in. by 11.25 in.) and shall be located on the cabinet top as in the GFE. The exhaust air outlet shall be provided with a bolt hole pattern as in the GFE for connecting to external ducting.

3.3.1.17.1.4 Air vane switch. The final power amplifier cabinet shall be equipped with an air vane switch that shall be activated when the main blower cooling air pressure is reduced below that produced by a safe airflow rate (600 CFM at sea level). The AM-6047B/FRT-84(V) shall be automatically disabled by activation of the air vane switch. The safe airflow rate shall be the airflow rate required to maintain the seal temperature of the AM-6047B/FRT-84(V) tube below the manufacturer's recommended maximum with the transmitter operating on electrical power with line voltage at a frequency of 47 Hz with an RF power output of 10 kW PEP (5 kW avg.) in any operating environmental condition specified in 3.8. Actuation of the air vane switch shall shut off the AM-6047B/FRT-84(V), causing a transmitter fault and illuminating the XMTR FAIL lamp on the exciter front panel and the fault lamp on the final power amplifier cabinet meter panel.

3.3.1.17.1.4.1 Restoration of air pressure. When the main blower cooling air pressure is restored to a safe airflow rate, the air vane switch shall be automatically reset to restore the transmitter to normal operation.

3.3.1.17.1.5 High temperature cut off switch. The final power amplifier cabinet shall be equipped with a high temperature cut off switch that monitors the cooling air exhaust temperature. The thermal sensor shall be manually adjustable to actuate the cut off switch at any temperature in the range of +65°C to +121°C inclusive. Actuation of the cut off switch shall cut off the AM-6047B/FRT-84(V) filament and bias supply, causing a transmitter fault and

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illuminating the XMTR FAIL lamp on the exciter front panel and the fault light on the final power amplifier cabinet meter panel. The final power amplifier cabinet blower shall continue to operate. The AM-6046B/FRT shall revert to the standby condition.

3.3.1.17.2 Thermal survey. When specified in the contract or order (see 6.2.1) the contractor shall conduct a complete thermal survey of the FAS prior to offering it for Government testing to determine the existence of hot spots which may prove to be injurious to the useful life of the transmitter.

3.3.1.18 Acoustic noise. The transmitter shall produce no more than 65 dB of acoustic noise measured at sea level by instrumentation having an A weighting characteristic in accordance with MIL-STD-1472 (A weighting, ANSI S1.4-1961) at a distance of 3 m (10 ft) in any direction normal to the nearest outer surface of the transmitter when it is installed and operated in a large transmitter room without special acoustic treatment and with an ambient noise level not greater than 50 dB. (Reference noise level is a free progressive 1000-Hz sound wave of 10^{-16} W per square centimeter (W/cm^2)). Acoustic treatment of the AM-6046B/FRT, the AM-6047B/FRT-84(V), and cabinet walls and doors is permitted if necessary to conform to this requirement.

3.3.1.19 Safety to personnel. Transmitter design shall conform to the Safety criteria paragraph of MIL-E-16400.

3.3.1.19.1 Interlocks. All transmitter compartments in which a potential of 50 V root-mean-square (rms) is exceeded shall be enclosed and protected with interlocks on all access doors or panels which may be opened during operation or maintenance. Interlocks shall act to remove all potentials within the protected compartment. Interlock voltage shall not exceed 50 V rms. In cases where the removal of a potential is impractical, such as line input terminals, the terminals shall be insulated and protected with a nonmetallic cover labeled CAUTION and stating the voltage.

3.3.1.19.1.1 Interlock circuit extension. A pair of terminals from which the interlock circuit may be extended for at least 61 m (200 ft), using the same sized conductors as used within the internal interlock circuit, shall be provided. Terminals shall be located in the bottom of the exciter-driver cabinet.

3.3.1.19.2 Capacitor discharge. Provisions shall be made to prevent access to any compartment of the transmitter without activation of a mechanical device to discharge any capacitor in which the product of the capacitance, in farads, and the square of the operating voltage, in V, exceeds 25, except that such device need not be provided for a compartment where the maximum voltage in the compartment does not exceed 50 V rms.

3.3.1.19.3 Final power amplifier cabinet shorting stick. An electrical shorting stick with readily accessible stowage and cable of sufficient length to reach all high-voltage capacitors anywhere in the transmitter shall be provided and stored in the final power amplifier cabinet. The cable shall be permanently connected to a ground point having less than 0.01-ohm resistance. The handle of the shorting stick shall withstand a voltage of at least 40,000 V for a minimum of 10 minutes without arcing, burning, or insulation breakdown.

3.3.1.19.4 AM-6046B/FRT shorting stick. An electrical shorting stick shall be provided and stored inside the AM-6046B/FRT. The shorting stick shall be accessible when the AM-6046B/FRT is withdrawn on its slide and shall be furnished with cable of sufficient length to reach all high-voltage capacitors in the AM-6046B/FRT. The cable shall be permanently connected to a ground point having less than 0.1-ohm resistance. The handle of the shorting stick shall withstand a voltage of at least 4000 V for a minimum of 10 minutes without arcing, burning, or insulation breakdown.

3.3.1.19.5 Overcurrent device location. A fuse or an overcurrent trip unit of a circuit breaker shall be connected in series with each ungrounded conductor as determined by the National Electrical Code (NFPA 70-1978) for grounded source power. The overcurrent device shall be connected to the load side of the power switch. If circuit breakers are used, they shall also serve as the main power switch. Circuit breakers shall provide a visible indication of status when tripped. (For the primary three-phase power disconnect (see 3.3.1.2.2).)

3.3.1.19.6 Weight marking. Components weighing in excess of 18 kilograms (kg) (39.6 pounds (lb)) shall have the weight clearly identified on the external surface and readily visible during installation and removal.

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3.3.1.20 Protection of internal components. All major components of the transmitter shall, upon failure, activate protective features preventing further damage. All electrical indicating instruments with associated multipliers or transformers, except self-contained thermocouple instruments, shall be protected, by the use of bypass units, against damage due to stray RF currents or transient potentials. All delicate instruments and devices shall be mounted for protection from damage due to shock and vibration. (See 3.2 for a list of major components.)

3.3.1.20.1 Protection from operator error. The design and construction shall be such that damage to the transmitter shall not result from operator error such as improper control sequencing, improper tuning, failure to respond to alarms or fault indicators, and failure to follow operating instructions.

3.3.1.20.2 External circuit fault protection. External circuit fault protection which conforms to the Electrical tests paragraph of MIL-E-16400 shall be provided. All components shall be protected from damage in the event of failure or malfunction of any external circuit, including prime power, audio input sources, output coaxial line, or other portions of the load circuit. Failures or malfunctions of external circuits shall include all possible types such as open circuit, short circuit, intermittent open and short circuits, excessive voltage-standing-wave-ratio (VSWR), long term overvoltage and undervoltage of prime power, improper phase rotation, loss of one or more phases, and transients.

3.3.1.20.3 VSWR protection. Operation into a load which generates a VSWR of 4.0:1 shall not cause damage. Provision shall be included for manual reset and for a manual calibrated adjustment of the VSWR sensing circuits to interrupt transmitter operation between the limits of 2.5:1 and 4.0:1. The VSWR protection system shall be capable of automatic adjustment of transmitter output impedance to match out variations in antenna impedance equivalent to a VSWR of 4.0:1 or less.

3.3.1.20.4 Internal transient voltage protection. Internal transient voltage protection circuitry shall be incorporated into the transmitter. The circuitry shall include, but not necessarily be limited to, protection from transient voltages generated by transmitter shut-downs, circuit breaker actuation, circuit breaker deactuation, change of frequency, changing modes of operation, proximity of cabling, sudden short or open in transmission output line, loss of main AC input power on any combination of phase or phases, any form of arcing, and any other transient voltage condition.

3.3.1.20.5 Power supply protection. Fault conditions ranging from open circuits to short circuits shall cause no damage to power supplies.

3.3.1.20.6 Printed circuit board (PCB) protection. With power applied to the transmitter, removal or insertion of PCBs shall not damage or degrade any part of the transmitter. Each PCB shall be keyed to prevent insertion into the wrong connector, and to prevent reversal of insertion.

3.3.1.21 Test points. The transmitter shall be provided with adequate and well-defined test points to facilitate making performance measurements and for use in maintaining the transmitter. Test points and test features shall be in accordance with the requirements of MIL-STD-415.

3.3.1.21.1 Test jacks. Test jacks provided in accordance with 3.3.1.21 and subject to the influence of interference shall be equipped so that shielded test leads can be used. When a test jack is to be used with a spectrum analyzer, the output of the test jack shall be within the range of -50 to 0 decibels referred to one milliwatt (dBm). All RF sampling voltages shall be 1 V rms based on a 10 kW PEP output when modulated with a two-tone signal. A jack shall be provided to insert the output of a two-tone oscillator at each audio input of the exciter.

3.3.1.22 Test cords and module extenders. All test cords, PCBs, or module extenders required for operation and maintenance of the transmitter shall be provided with the equipment. Signals and power for servicing and testing of modules in the extended position shall be provided by the basic unit.

3.3.1.23 General purpose electronic test equipment (GPETE). The transmitter shall be designed to make maximum use of GPETE selected from MIL-STD-1364. GPETE referenced in all technical documentation, including manuals and training aids, shall be the same as that specified in the manuals provided as Government-furnished material (GFM) (see 6.5). If the Government does not provide manuals, GPETE shall be selected from MIL-STD-1364.

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3.3.1.24 Special purpose electronic test equipment (SPETE). The design of the transmitter shall limit requirements for special tools (see 6.6) and SPETE to a minimum. SPETE and special tools, if required and when specified in the contract or order (see 6.2.1), shall be provided with each transmitter.

3.3.1.25 Meters and indicators. Meters shall be provided to indicate all critical circuit parameters where the function indicated is of the continuously variable type, such as plate current, output power, bias voltage, and exciter output. Unless otherwise specified herein, meters shall be of an accuracy of 5 percent or better.

3.3.1.25.1 Multi-test meters. Multi-test meters shall be provided in the exciter, the KY-655A/FRT, the AM-6046B/FRT, and the AM-6047B/FRT-84(V) and shall be capable of being switched to any circuit which does not require continuous monitoring but requires periodic adjustment or monitoring inspection.

3.3.1.26 Controls. All transmitter operational controls shall be located on the front panels and be so grouped, arranged, and integrated by function with the applicable meters and indicators that an orderly, simple, and efficient method of performing normal operational functions may be carried out by a single operator.

3.3.1.26.1 Control markings. Words or phrases descriptive of the control functions shall be marked adjacent to each operating control. Markings shall be permanent and shall be easily readable by a person with normal vision at a distance of 60 cm (23.6 in.) with an illumination as low as 11 lux (one foot-candle).

3.3.1.27 Wiring connections. All wiring connections for power, signals and control required for operation of the transmitter shall be located on the rear of each unit case or chassis assembly, as applicable, except the transmitter RF output point shall be located on the top of the final power amplifier cabinet and external signal lines shall connect directly to mating connectors on the rear of the exciter and the keyer.

3.3.2 Exciter design requirements. The exciter shall contain all of the output frequency determining elements when operated on the internal frequency standard and all of the elements determining the output modulation characteristics sufficient to conform to the mode requirements specified in 3.7.1.2. The exciter shall provide the termination for the audio input lines and shall furnish to the AM-6046B/FRT a modulated RF excitation signal, with output power adjustable between a nominal minimum of not more than 10 mW PEP and a maximum of not less than 250 mW PEP, across 50-ohms nominal impedance in any of the 280,000 possible operating frequencies within the frequency range of 2.0000 MHz to 29.9999 MHz in 100 Hz steps. The exciter shall provide to the AM-6046B/FRT and the AM-6047B/FRT-84(V) all required tuning and conditioning information. The exciter shall contain all of the input audio line terminations, the remote control interface and remote control data signal and processing circuitry, and the input signal processing devices and circuitry. The exciter shall provide highly reliable and stable operation consistent with the conditions specified herein. The exciter shall be frequency-synthesized, having self-contained provisions for accepting modulating signals from one, two, three, or four 3-kilohertz (kHz) bandwidth audio input channels. The exciter signal conversion scheme shall use nominal 1.75 MHz sideband carrier modulators whose outputs are up-converted to a nominal 112 MHz for further amplification and filtering, followed by down-conversion to the final output frequency range of 2.0000 MHz to 29.9999 MHz and broadband amplification to the desired output level. The exciter output shall be capable of a maximum of not less than 250 mW PEP RF power in an A3J, A9B, or A9J operating mode depending upon the number of audio frequency inputs utilized. The exciter shall conform to the stability requirements specified in 3.7.2.1 when terminated into a nominal 50-ohm load with a VSWR not greater than 1.5:1.

3.3.2.1 Exciter configuration. Unless otherwise specified in the contract or order (see 6.2.1), the exciter shall be configured for 2 ISB operation on channels A_1 and B_1. In the 2 ISB configuration, the side carrier generator simulator module A20 shall be installed in place of the side carrier generator required for 4 ISB operation and the sideband filters and modulator modules for channels A_2 and B_2 shall not be furnished. Only the replacement of the side carrier generator simulator and installation of sideband filters and modulator modules for channels A_2 and B_2 shall be required for a change in configuration from 2 ISB to 4 ISB.

3.3.2.2 Exciter primary power. The exciter shall operate from a single-phase power source of 115 V +10 percent and 230 V +10 percent, 47 Hz to 63 Hz. Conversion from operation at 115 V to 230 V, or vice versa, shall be accomplished by a single toggle switch located in the exciter power supply module.

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3.3.2.3 Frequency selection. The operating frequency shall be selectable by the RCS and by manual controls on the front panel of the exciter. The absolute frequency of the RF carrier, whether present in the transmitter output or fully suppressed, shall be determined by the exciter and selectable by manual controls on the front panel of the exciter or by the RCS when so equipped.

3.3.2.4 Exciter synthesizer module. The synthesizer module shall use the 3 MHz and 30 MHz outputs from the auxiliary frequency generator module and the binary coded outputs from the front panel frequency selector switches to generate the 82 MHz to 110 MHz output frequencies required by the down-converter module for final down-conversion to the 2.0000 MHz to 29.9999 MHz exciter output frequencies. The front panel frequency selector switch binary code shall be converted to binary-coded-decimal (BCD) code (to permit the use of standard BCD code programmable divider integrated circuits (IC) in the phase locked loop circuits) with a complementary metal-oxide semiconductor (CMOS) fusible link programmable read-only memory (PROM) IC. A phase locked loop, with oscillator output frequencies of 28.000 MHz to 28.999 MHz in 1 kHz steps as determined by the 10 kHz, 1 kHz, and 100 Hz frequency selector controls, shall be translated to 117.2 MHz to 117.1001 MHz by first being divided by 10 (in order to obtain the 100 Hz frequency change steps of the exciter) to 2.8000 MHz to 2.8999 MHz and then mixed with 120 MHz generated from the 30 MHz signal in a times-4 multiplier. The 117.2 MHz to 117.1001 MHz output shall be mixed into a phase lock loop whose voltage-controlled oscillator (VCO) covers the desired synthesizer output of 82.0001 MHz to 110.0 MHz and is also controlled in 100 kHz steps by the 10 MHz, 1 MHz, and 100 kHz frequency selector controls. The 100 kHz and 1 kHz reference frequencies required by the phase detectors in the two-phase-locked loops shall be obtained from the 10 kHz and 100 kHz outputs derived from dividing down the 30 MHz input in the reference divider circuits. A prescaler IC shall be used in the programmable divider of the 28 MHz phase locked loop in order to achieve reliable frequency division in 1 kHz steps at 28 MHz.

3.3.2.4.1 Auxiliary frequency generator. The 30-MHz phase-locked loop PCB shall use three low power Schottky-clamped transistor-transistor-logic (TTL) ICs (a quadruple 2-input NAND gate, a decade counter, and a dual J-K flip-flop) in the frequency divider, phase detector, and buffer circuits for low component count, low power dissipation, and higher reliability.

3.3.2.4.2 1.75/113.75 MHz generator. The 113.75 MHz PCB shall use a single dual J-K flip-flop low power Schottky TTL IC for the divide-by-four frequency divider in the phase locked loop, hermetically sealed dual balanced mixers for the phase detector and the 6.25 MHz translation mixer, temperature-compensated 113.75 MHz VCO, and loop gain sufficient to ensure against loss of phase lock over the entire operating temperature range of the exciter.

3.3.2.5 Transmitter gain control (TGC) module. The TGC module shall be completely electronic with no motor drives, relays, or mechanically adjusted limit switches. The control voltage generated by the TGC module and supplied to the down-converter module to control the exciter RF output shall be derived by comparison of the TGC voltage from the AM-6047B/FRT-84(V) stages with the reference voltage set by an adjustable potentiometer for the desired RF output level. The output of the comparator shall gate an up and down counter driving an analog-to-digital converter to produce the control voltage.

3.3.2.6 Down-converter module. The down-converter module shall use feedback stabilized wideband amplifiers requiring no tuning adjustments, and pin-diode gain control circuits for improved stability and reliability.

3.3.2.7 Output amplifier module. The output amplifier module shall be capable of withstanding a direct open or short circuit on the output while operating without causing damage. A hybrid amplifier output stage and a feedback stabilized preamplifier stage shall be used for improved efficiency in order to reduce the input power requirement and reduce the temperature rise.

3.3.2.8 Exciter power supply. The exciter power supply shall be configured to enhance the flow of cooling air. The power supply shall contain short circuit protection and overvoltage protection of the + 24 V and + 5 V regulators to protect the frequency standard oven on the +24 V circuit and the ICs operating from the +5 V circuit.

3.3.2.9 Frequency standards. The frequency stability of the transmitter shall be determined by the exciter when operating on the 1-MHz internal frequency standard.

3.3.2.9.1 Internal frequency standard. The 1-MHz internal frequency standard shall have stability equal to that specified in 3.7.2.1. The internal frequency standard shall provide three separate 1-MHz outputs at a level of 1 V rms when terminated into a 50-ohm load. Two of

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these three outputs shall be brought out to connectors on the front and rear panels of the exciter. The third output shall feed the synthesizer circuitry.

3.3.2.9.1.1 Rear panel connector. One coaxial connector of 50-ohms impedance marked 1 MHz INT STD MON shall be mounted on the rear panel of the exciter. The connector shall mate with a standard 50-ohm Type BNC fitting.

3.3.2.9.1.2 Front panel connector. One coaxial connector of 50-ohms impedance marked 1 MHz INT STD MON shall be mounted on the front panel of the exciter. The connector shall mate with a standard 50-ohm Type BNC fitting.

3.3.2.9.1.3 Internal frequency standard oven. The 1-MHz internal frequency standard shall be equipped with a +75°C, solid-state proportional controlled, crystal oven selected in accordance with MIL-O-39021. Power for the oven shall be 24 V direct current (VDC) from the exciter power supply and shall remain on when primary power is applied to the exciter. When the internal frequency standard oven is operating, an exciter front panel green lamp labeled STD OVEN shall be illuminated. If the oven fails, the lamp shall be extinguished.

3.3.2.9.2 External frequency standard. The exciter shall be provided with a means of connecting to an external 1-MHz prime frequency standard in lieu of the internal frequency standard. The exciter shall conform to performance requirements when using the external standard as long as the external standard provides a signal equal to, or better than, the internal standard with respect to stability, amplitude, and purity. Connectors which mate with a standard 50-ohm Type BNC fitting shall be located on both the front and rear panels of the exciter to receive the 1-MHz external frequency standard signal. The level of this signal shall be 1 V rms across a 50-ohm load. The front panel connector shall be marked 1 MHz EXT STD MON; the rear panel connector shall be marked 1 MHz STD IN.

3.3.2.9.2.1 Frequency standard synchronization. The exciter shall be equipped with a manually adjustable circuit and a control located on the auxiliary control panel for synchronizing the internal frequency standard with the 1-MHz external frequency standard. If the external standard fails, the internal standard shall automatically determine the exciter output frequency, stability, and accuracy. When the circuit test selector specified in 3.3.2.19.16 is in the FREQ STD LOCK position, synchronization shall be accomplished by manual adjustment of an internal control.

3.3.2.9.2.2 Switching. A failure sensing modular device shall be provided to switch frequency control from the external to the internal standard upon failure of the external standard. Automatic switching shall occur when the amplitude of the external standard decreases 3 dB below a 1 V rms reference level. After restoration of the external standard to at least 1 V rms, the sensing device shall automatically switch back to the external frequency standard. The allowed transit time for the switching device to function shall not exceed 10 milliseconds (ms). No degradation to the transmitter output shall occur as a result of the 3 dB amplitude decrease at the output of the external standard.

3.3.2.9.2.3 Standard fail indicator. A front-mounted visual indicator with provisions for remoting to other locations shall be provided to indicate the loss of external frequency standard output. When the internal frequency standard is in use, the indicator, labeled STD FAIL, shall be illuminated in red. When the external standard is in use, the STD FAIL lamp shall remain lighted.

3.3.2.10 Voice frequency gate (VFG). Provision shall be made in the exciter, audio input channel A_1, for a VFG circuit equipped with an internal adjustment control on the VFG module for setting the gate sensitivity to any signal sensing level ranging from -5 dBm to -20 dBm.

3.3.2.10.1 VFG keying line. The keying line for VFG operation shall operate in conjunction with the exciter main internal keying circuit for actual exciter keying purposes.

3.3.2.10.2 VFG hold time adjustment. A VFG hold time control to provide for continuous adjustment of VFG hold time between 100 ms and 3 seconds (s) shall be mounted on the auxiliary control panel internal to the exciter.

3.3.2.11 External push-to-talk (PTT) service. External PTT service shall be selectable at the exciter front panel. When external PTT service is in use, the audio signals shall be fed to the input of channel A_1.

3.3.2.12 Automatic level control. The audio inputs shall be arranged so that each input signal, at any level above 0 dBm, shall automatically and independently be attenuated in a

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smooth transition to the point where a +10 dBm input signal level shall not exceed a +3.0 dBm increase in the signal level to the AM-6046B/FRT. (The AM-6046B/FRT and the AM-6047B/FRT-84(V) shall be similarly automatically protected from high signal levels that would otherwise exceed designed power limits of the transmitter and cause a shutdown.)

3.3.2.13 Dry contact keying. A pair of terminals and associated circuitry shall be provided in the exciter for dry contact ON-OFF or external PTT keying in the USB mode of the exciter and the AM-6047B/FRT-84(V) key-up noise reduction circuitry during simplex operation.

3.3.2.14 Transmitter status read-back function. Provision shall be made in the exciter for sensing, monitoring, and translating transmitter conditions to dry-contact control of a remote readout indicating system. The remote readout function shall be as follows:

Modulation channel	Condition	Contacts
A_1	On	Closed
B_1	On	Closed
A_2 and B_2	On	Closed
Parameter sensed	**Condition**	**Contacts**
B+	On	Closed
RF	On	Closed

3.3.2.15 Front panel bulb voltage. Exciter front panel bulbs shall be operated from a source voltage which is reduced 35 percent below the nominal voltage rating of the bulbs.

3.3.2.16 Exciter and KY-655A/FRT interface. The exciter and the KY-655A/FRT shall be provided with interface connectors so that when the A1, F1, or F4 class of emission is selected at the exciter, the KY-655A/FRT is automatically placed in operation (see 3.3.2.19.2.2).

3.3.2.17 Exciter standby condition. Provisions shall be made for an exciter standby condition. For definition of exciter standby condition, see 6.4.5.

3.3.2.18 Two ISB simplex. The exciter shall be capable of simplex operation when the 2 ISB mode is selected for use. The incorporation of a simplex operational capability in the 2 ISB mode shall not degrade performance characteristics of the transmitter when operating in any of the modes specified herein.

3.3.2.19 Exciter controls, control-indicators, indicators, and meters. In addition to controls, control-indicators, indicators, and meters specified in 3.3.2.19.1 through 3.3.2.19.16, the exciter shall be equipped with controls, control-indicators, and meters specified in 3.3.2.19.1 through 3.3.2.19.17.3.

3.3.2.19.1 Frequency controls. The control and display of the selected frequency shall be functions of one device. The selected frequency display shall be located on the exciter front panel, shall be illuminated from the rear and shall use Arabic numerals to indicate which of its possible digits has been selected. The controls shall be constructed so that there is no possibility of the exciter carrier frequency being other than that which is indicated. The size of the numerals and their illumination shall be such that they may be easily read by a person with normal vision at a distance of 60 cm (23.6 in.) when illuminated at a level of 11 lux (one foot-candle).

3.3.2.19.1.1 Number and function of controls. There shall be one digital, decimal control for each digit of the frequency to be selected. This requires six controls, one for each of the following:

a. Tens of MHz

b. Units of MHz

c. Hundreds of kHz

d. Tens of kHz

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e. Units of kHz

f. Tenths of kHz

These controls shall be arranged in a horizontal line reading from left to right in conventional form. A decimal point shall be located between the units of kHz and tenths of kHz controls.

3.3.2.19.2 Operating mode selection. The exciter shall provide modes of emission as specified in 3.7.1.2. To accomplish this requirement a control marked SIDEBAND SELECTOR and a control marked CLASS OF EMISSION shall be provided on the exciter front panel. These controls shall be in the form of solenoid driven rotary selector switches.

3.3.2.19.2.1 Sideband selector control. The sideband selector control shall provide selection of the following sideband operating combinations:

a. USB (channel A_1) with VFG

b. USB (channel A_1) with PTT

c. USB (channel A_1) continuously keyed

d. Lower sideband (LSB) (channel B_1) continuously keyed

e. Two channels ISB (channels A_1 and B_1) continuously keyed, except as specified in 3.7.2.3.1.1

f. Four channels ISB (channels A_2, A_1, B_1, and B_2) continuously keyed

3.3.2.19.2.2 Class of emission control. The class of emission control shall provide carrier insertion levels and emission capabilities as follows:

a. AØ operation, continuous carrier insertion at rated average power output, no modulation will be accepted

b. A1, F1, or F4 operation, providing the capability of the exciter accepting these forms of modulation in the A_1 channel

c. A2 or A3 operation, this function shall be interlocked with sideband selector control to provide operation with modulation in the A_1 channel and carrier insertion 6 dB below rated PEP output

d. Carrier insertion 10 dB below rated PEP output

e. Carrier insertion 20 dB below rated PEP output

f. Carrier insertion 40 dB below rated PEP output

g. Fully suppressed carrier (at least 60 dB below rated PEP output)

3.3.2.19.2.3 Simplex 2 ISB selector. A two-position internal switch shall be provided on the auxiliary control panel of the exciter for selecting normal or simplex 2 ISB operation. One position of the switch shall be marked NORMAL and the other position shall be marked SIMPLEX 2 ISB. When the switch is placed in the NORMAL position and the 2 ISB mode is selected at the sideband selector, the transmitter shall provide key-down operation in the 2 ISB mode as in the GFE. When the switch is placed in the SIMPLEX 2 ISB position and the 2 ISB mode is selected at the sideband selector, the transmitter shall provide that type of operation without the necessity of any other form of conditioning.

3.3.2.19.3 Individual subchannel gain. Each of the audio input channels shall be provided with a single front panel control which controls the amount of output power in its respective portion of the output spectrum. It shall be possible, through the use of these controls, to vary from 5 percent to 85 percent of the available output power in any one of the audio channels. Thus, it shall be possible, by using these controls to adjust the 1.75 MHz intermediate frequency (IF) circuitry, to have any combination of power division (totalling not more than 100 percent) among the four (or two) channels between the limits of 5 percent minimum and 85 percent maximum in any one channel. Controls shall be continuous and shall be calibrated to show every 5 percent from 0 percent to 70 percent, and every 10 percent from 70 percent to 100

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percent. Accuracy or calibration shall be 5 percent of the maximum value as measured in the transmitter output power.

3.3.2.19.4 Exciter power output control. The power output level of the exciter shall be variable from a nominal 10 mW to at least 250 mW by means of a single control while retaining all of the performance characteristics specified herein. The maximum output level when in the minimum position shall be not greater than 10 mW. The control shall be located on the auxiliary control panel in back of the front panel for ready accessibility when the exciter is extended for maintenance or adjustment.

3.3.2.19.5 Amplifier OFF. The amplifier OFF control-indicator, illuminated in white when the AM-6047B/FRT-84(V) is off, shall turn the AM-6047B/FRT-84(V) amplifier off. The indicator shall show the actual status of the transmitter. When the amplifier OFF control-indicator is actuated, only the internal frequency standard oven in the exciter shall remain on.

3.3.2.19.6 Standby. The standby control-indicator, illuminated in white, shall provide a command for the transmitter to go to a standby condition. The indicator shall show the actual status of the transmitter. Positive visual indication of both standby and full operation shall be provided on the front panel of the exciter. Full deactivation shall be indicated by an off indication for both conditions.

3.3.2.19.7 Operate. The operate control-indicator shall be illuminated in green to show transmitter operate condition. The control shall command the transmitter to a full operate (all voltages applied) condition.

3.3.2.19.8 Tune. The tune control-indicator shall be illuminated in white to indicate the transmitter needs to be retuned, for example, when a change in frequency or class of emission is selected. Operator actuation of the control shall initiate a tune sequence in the transmitter and automatically provide an operate command upon proper completion of the tune cycle.

3.3.2.19.9 Ready. The ready indicator shall be illuminated in green to indicate the transmitter is in operate, has completed a tune sequence, and is ready for the beginning of data transmission.

3.3.2.19.10 Local and remote operation selector. A local and remote selector shall be provided on the exciter. When the selector is in the remote position and an LCU is installed, it shall be possible to remotely perform frequency selection, class of emission selection, sideband selection, and transmitter status changes from standby to operate and from operate to standby. In the local position the data outputs from the LCU shall be inhibited and system control shall be done from the front panel of the exciter. The local control condition shall be encoded by the LCU for notification of the remote control unit (RCU) by means of the monitor message reply word. Status information, with the exception of frequency, emission, and mode, shall be contained in the monitor message reply word when the system is in the local control mode.

3.3.2.19.11 Internal frequency standard calibration. A control shall be provided for adjustment of the frequency of the internal frequency standard of the exciter. The control shall have an adjustment range sufficient to allow reduction of the absolute frequency error as necessary once every 90 days for a period of at least 10 years. The control, located on the auxiliary control panel, shall be accessible for use while the exciter is in service but shall be so arranged or protected that inadvertent movement of any control is unlikely. The control shall have sufficient vernier spread to allow the minimum frequency change to be small enough to reduce the absolute frequency error of the output carrier frequency to nominal zero error when the exciter is operating at any frequency between 2.0000 MHz and 29.9999 MHz. The control shall have a scale or other marking which allows a log to be obtained for long term calibration record purposes. For definition of frequency calibration of the exciter, see 6.4.4.

3.3.2.19.11.1 Calibration reference signal. The internal frequency standard signal present at the coaxial output specified in 3.3.2.9.1.2 may be used for calibration. Any absolute frequency error in the exciter output frequency shall have a simple arithmetic relationship to the absolute frequency error of the 1-MHz output signal of the frequency standard in use.

3.3.2.19.12 Input level controls. Four input level controls, one for each of the four channels, shall be located on the auxiliary control panel (see 3.3.2.19.17). By means of these controls it shall be possible to adjust the input level of each channel between the limits of +10 dBm to -30 dBm.

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3.3.2.19.13 Exciter failure indicator. A red, self-illuminated indicator labeled EXC FAIL shall be mounted on the exciter front panel. The indicator shall be illuminated when a fault occurs in the exciter circuits. Absence of a direct current (DC) operating potential or an RF injection frequency, or exciter overheating, shall light the EXC FAIL lamp.

3.3.2.19.14 Transmitter failure indicator. A red, self-illuminated indicator labeled XMTR FAIL shall be mounted on the exciter front panel. The indicator shall be illuminated when a fault occurs in the overall transmitter. The XMTR FAIL lamp shall be illuminated when the EXC FAIL lamp is lighted. Illumination of the XMTR FAIL indicator shall indicate one or more of the following faults:

a. Faulty keyer (A1, F1, F4 modes only)

b. AM-6046B/FRT faults:

1. Excessive RF at final amplifier plate

2. Excessive final amplifier plate and screen current

3. Excessive temperature in the PP-6067A/FRT

c. AM-6047B/FRT-84(V) faults:

1. Excessive screen current

2. Excessive plate current

3. Excessive reflected power

4. Tuning time over 15 s

5. Variable capacitor at end stop limit

6. Shorting stick not hung up properly

7. Chassis not secured in cabinet

8. Cabinet bottom access door not secured

9. Plate compartment cover not secured

10. Input network compartment cover not secured

3.3.2.19.15 Volume unit (VU) input meter. One VU meter on the exciter front panel and an associated selector switch shall provide for individually checking each of the four audio input circuits. The meter shall electrically follow the attenuators (see 3.3.2.12) and be arranged so that the meter indicates 0 dBm when the attenuator is in the 0 dBm attenuation position and a 0 dBm 1-kHz signal is applied to the exciter audio input terminal. The meter shall have the same performance characteristics as the VU input meter in the exciter provided as GFE. A control on the auxiliary control panel shall provide for zero adjustment of the VU meter.

3.3.2.19.16 Exciter circuit test meter. The exciter circuit test meter shall provide for checking circuit parameters by positioning the circuit test selector and observing the meter indication. Selectable positions shall be arranged as follows when the selector switch is rotated in a clockwise direction:

a. A red sector, corresponding to the red sector of the meter, shall contain positions labeled:

1. +5 V

2. +15 V

3. +24 V

4. -12 V

5. +125 V

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b. Channel B_2 automatic level control (ALC)

c. Channel B_1 - ALC

d. Channel A_1 - ALC

e. Channel A_2 - ALC

f. Peak power control (PPC) and avg power control (APC)

g. TGC

h. A green sector, corresponding to the green sector of the meter, shall contain positions labeled:

1. 1.74371 MHz (channel A_2 sidetone)
2. 1.75 MHz (channels A_1 and B_1 sidetone)
3. 1.75629 MHz (channel B_2 sidetone)
4. 113.75 MHz (up-converter input)
5. 82-110 MHz (down-converter injection)

i. Power output (exciter RF output)

j. Frequency standard lock (see 3.3.2.19.16.1)

3.3.2.19.16.1 Frequency standard lock position. The circuit test selector position labeled FREQ STD LOCK shall permit manual adjustment of the internal frequency standard to the frequency of the external frequency standard by means of the internal frequency standard calibration control on the auxiliary control panel. One beat of the meter in 30 s shall indicate a frequency correlation of 0.3 Hz.

3.3.2.19.17 Exciter auxiliary control panel. The exciter auxiliary control panel shall be accessible when the exciter is withdrawn on its slide. In addition to the controls specified in 3.3.2.19.17.1 through 3.3.2.19.17.3, the auxiliary control panel shall be equipped with a modulation ON and OFF selector, a test A/B selector, and a normal/fault override selector for maintenance and installation testing.

3.3.2.19.17.1 Modulation ON and OFF selector. In the OFF position the selector shall remove modulation from the exciter output. When the selector is in the OFF position, the EXC FAIL indicator shall be illuminated.

3.3.2.19.17.2 Test A/B selector. The test A/B selector shall be used only for initial exciter installation to activate the exciter independent of other system units. Selection of the A position shall ground ready power. Selection of B shall ground readback power.

3.3.2.19.17.3 Normal/fault override selector. The normal/fault override selector shall be for use in troubleshooting to override the exciter fault circuit. The selector shall reset the exciter fault circuit when momentarily placed in the fault override position.

3.3.3 KY-655A/FRT design requirements. The KY-655A/FRT, hereinafter referred to as the electronic keyer, shall be a separate unit located within the exciter-driver cabinet and shall be capable of being easily removed and installed at a remote location. When removed for operation from a remote location, it shall be capable of being rack-mounted on a standard 48.3 cm (19 in.) rack in accordance with MIL-STD-189 and connected to the exciter by a single multi-wire cable.

3.3.3.1 Electronic keyer primary power. The electronic keyer shall operate from single-phase power sources of 115 VAC ±10 percent and 230 VAC ±10 percent, 47 Hz to 63 Hz. Conversion from operation at 115 VAC to 230 VAC, or vice versa, shall be accomplished by a single internal toggle switch protected by a guard to prevent inadvertent switching.

3.3.3.2 CW, frequency shift and analog keying. CW keying, single channel FSK and FAX (analog) keying shall be accomplished by means of the electronic keyer which shall be capable

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of accepting CW keying at speeds up to 200 bauds, FSK up to 300 bauds, and an analog input at speeds up to 400 bauds.

3.3.3.3 CW mode. The electronic keyer shall be designed so that CW operation is possible using either 60 milliamperes (mA), 20 mA, 100 V, ±6 V polar or dry contact input signals.

3.3.3.4 DC isolation. The electronic keyer shall be designed so that the DC input loop in all modes of operation shall be isolated from ground.

3.3.3.5 Electronic keyer and exciter interface. The electronic keyer and the exciter shall be provided with an interface that allows the electronic keyer to be connected to, and controlled by, the exciter. When so connected, selection of the A1, F1, or F4 class of emission shall automatically place the electronic keyer in operation. Screw type terminal strips with appropriate connections shall be available on the main chassis within the electronic keyer.

3.3.3.5.1 Automatic line transfer. When operating mode A1, F1, or F4 is selected, the electronic keyer output shall be automatically terminated into the input to channel A_1 of the exciter.

3.3.3.6 Electronic keyer controls. In addition to controls specified in 3.3.2.19.17 through 3.3.2.19.17.3, the electronic keyer shall be equipped with controls specified in 3.3.3.6.1 and 3.3.3.6.2.

3.3.3.6.1 Front panel controls. The electronic keyer shall include the following controls: a) a front panel switch shall allow operator selection of teletypewriter (TTY), CW, or FAX (normal or reversed) modes of operation; b) a direct reading 10-turn TTY deviation control shall allow operator selection of deviations from 1 Hz to 999 Hz about the selected center frequency; c) a front panel function test selector switch shall be provided to allow operator selection of TTY mark, TTY space, CW, FAX white, and FAX black for test purposes; and the controls specified in 3.3.3.6.1.1 through 3.3.3.6.1.3. When the function test selector is in a position other than operate, a front panel test indicator shall be illuminated (amber).

3.3.3.6.1.1 Electronic keyer center frequency selector. A four-position electronic keyer center frequency selector on the front panel shall provide a means of selecting frequency-shift tones, provided by the electronic keyer for teletype and facsimile operation, about the following center frequencies:

a. 2550 Hz

b. 2000 Hz

c. 1900 Hz

d. Adjustable from 400 Hz to 3000 Hz by operation of a FREQ ADJ control to set the center frequency when the center frequency selector is in the blank position.

3.3.3.6.1.2 Circuit test meter and switch. The electronic keyer circuit test meter and switch shall provide for checking circuit parameters by positioning the switch and observing the meter indication. Selectable positions shall be arranged as follows:

a. A green sector, corresponding to the green sector of the meter, shall contain positions labeled:

1. +10 V

2. -10 V

3. +18 V

4. -18 V

5. OUTPUT

b. OVEN

c. TTY

d. FAX

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3.3.3.6.1.3 Control-indicators. Three control indicators labeled TEST, STANDBY, and OPERATE shall provide for control of the electronic keyer. Control indicators shall be illuminated to indicate the status of the electronic keyer. The electronic keyer shall normally be automatically controlled by the exciter and the STANDBY and OPERATE controls shall by inoperable.

3.3.3.6.2 Electronic keyer internal controls. The electronic keyer shall be equipped with the internal controls specified in 3.3.3.6.2.1 through 3.3.3.6.2.3. Electronic keyer internal controls shall be accessible when the keyer is withdrawn on its slide.

3.3.3.6.2.1 TTY input selector switch. The TTY input selector switch shall provide for selection of the following keying loops (see 3.7.3.4):

a. 60 mA

b. 20 mA

c. 100 V

d. 50 V

e. Polar

f. Dry contact

3.3.3.6.2.2 Level control. The level control shall provide a means of adjusting the output level between a minimum of -10 dBm and a maximum of +3 dBm.

3.3.3.6.2.3 Delay control. The delay control shall provide a means of adjusting the transition relay delay time between a minimum of 100 ms and a maximum of 5 s.

3.3.4 AM-6046B/FRT design requirements. The AM-6046B/FRT, hereinafter referred to as the driver amplifier, shall be a two-stage (driver amplifier stage and power amplifier stage), linear amplifier that produces a nominal output of 800 W PEP or 400 W avg power with a nominal input of not greater than 250 mW. The driver amplifier shall consist of two stages: a driver amplifier stage using two Type 8122 tubes and a power amplifier stage using two Type 4CX1500 tubes, EIMAC, or equal.

3.3.4.1 Driver amplifier primary power. The PP-6067A/FRT power supply shall provide primary power for the driver amplifier.

3.3.4.2 Screen power supply protection. The driver amplifier screen power supply shall be protected from damage due to a plate-to-screen short in any tube used in the driver amplifier. The 10 W Zener diodes in the screen Zener string (assembly 1A1A9) shall be Type UZ 7840, UZ 7860, and UZ 7875, or equivalent.

3.3.4.3 Filament low voltage transformer circuit protection. The primary circuit of the driver amplifier filament low voltage transformer (1A1T1) shall be fused.

3.3.4.4 Driver amplifier final assembly. The driver amplifier final assembly shall include 19 primary frequency bands as follows:

a. Four bands of 0.5 MHz bandwidth from 2.0 MHz to 4.0 MHz

b. Four bands of 1.0 MHz bandwidth from 4.0 MHz to 8.0 MHz

c. Eleven bands of 2.0 MHz bandwidth from 8.0 MHz to 30.0 MHz

3.3.4.4.1 Transformer assembly rotary switch. The driver amplifier final transformer assembly shall employ a 24-position rotary multiple selector switch to short out primary and secondary coil assemblies associated with the 19 operating frequency band codes generated by the band repeater assembly. This shorting out provision shall eliminate ringing resulting from open-circuit self-resonance and consequent corona and arcing. The switch shall be constructed so that all rotor contacts rest on stator buttons, not on the base insulation material, when the switch is in any of the 24 positions. Contact buttons shall be chamfered to remove any corona inducing sharp edges. Physical separation of the RF current carrying bifurcated rotor fingers shall be incorporated to ensure contact by both members. The switch shall rotate in one direction only and shall have no detents or stops. Each of the seven rotor contacts and each of

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the 30 shorting fingers shall align with its related stator contact within ±1 degree of arc. Base insulation material used in the switch shall withstand a voltage of 8000 V peak, ±5 percent, AC and DC, without breakdown.

3.3.4.5 Tuning method. The driver amplifier shall be automatically tuned without RF energy in the 19 frequency bands specified in 3.3.4.4. Frequency information shall be conveyed by a five-wire binary code supplied by the exciter through the band repeater assembly in the final power amplifier cabinet. Frequency tuning of the driver amplifier shall be accomplished automatically when a frequency is selected and the tune control on the exciter is depressed in local operation. Frequency tuning under remote control shall be automatic.

3.3.4.5.1 Manual tuning methods.

3.3.4.5.1.1 Electro-manual tuning. Electro-manual tuning of the driver amplifier shall be possible by manual rotation of the driver amplifier front panel tuning control to select any of the 19 frequency bands. The window above the tuning control shall display the frequency of the selected band. Positioning the tuning control shall cause the driver amplifier to be tuned electrically to the selected band.

3.3.4.5.1.2 Full manual tuning. It shall be possible to accomplish full manual tuning of the driver amplifier by withdrawing the driver amplifier on its slide and physically rotating the final transformer asssembly (1A1A2) and the driver transformer assembly to the desired frequency band.

3.3.4.6 Driver amplifier input and output impedances. The driver amplifier input and output impedances shall be 50-ohms nominal. Throughout the frequency range of 2.0000 MHz through 29.9999 MHz, the input circuit impedance shall be 50-ohms nominal resistive with reactive components reduced to values which will cause a VSWR no higher than 1.5:1 at the exciter output.

3.3.4.7 Driver amplifier standby. The driver amplifier standby condition is identical to the exciter standby condition specified in 6.4.5 with the following addition: All components and circuitry in the driver amplifier requiring more than 1 s to reach full operating condition shall remain activated during standby. All circuitry remaining active during the standby condition shall be protected so that the cabinet may be opened for maintenance without violating any personnel safety requirements (see 3.3.1.19).

3.3.4.8 Output coupling capacitor 1A1C4. The driver amplifier RF output coupling capacitor 1A1C4 shall have a rating of not less than 6000 V at 125°C and 12 A at 30 MHz and 125°C.

3.3.4.9 Screen and plate supply rectifiers. Type JANTX 1N5622 high-voltage rectifiers in accordance with MIL-S-19500/427 and with maximum ratings of 1000 V peak inverse voltage, 900 mA at 75°C, and surge current of 50 A shall be used in the 600 V driver amplifier screen and plate supply.

3.3.4.10. Driver amplifier controls, meters, and indicators. Driver amplifier controls, meters, and indicators shall conform to those in the GFE.

3.3.4.10.1 Driver amplifier output power level control. The driver amplifier output power level control on the driver amplifier front panel shall permit operator adjustment of the driver amplifier output power between the limits of full rated power and 25 percent of full rated power.

3.3.4.10.2 Bias control. The bias control on the front panel of the driver amplifier shall permit screwdriver adjustment of the driver stage bias over the normal Class A operating bias for the driver stage tubes (0 VDC to -24 VDC).

3.3.4.10.3 Driver amplifier multipurpose meter and selector. The driver amplifier multipurpose meter and selector shall provide for checking circuit parameters by positioning the switch and observing the meter indication. Selectable switch positions shall include:

a. Power amplifier tube 1 plate current

b. Power amplifier tube 2 plate current

c. Driver amplifier tube 1 plate current

d. Driver amplifier tube 2 plate current

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e. Input power (0 mW to 150 mW)

f. Power amplifier plate voltage

g. Driver amplifier plate voltage

h. Power amplifier screen voltage

i. Driver amplifier screen voltage

3.3.4.10.4 Overload indicator. An overload indicator lamp on the driver amplifier front panel shall be illuminated in red to indicate final tube plate current in excess of 1 A and excess RF voltage at the plate of the final tubes.

3.3.4.10.4.1 Overload reset switch. An overload reset switch with positions labeled OVERLOAD RESET and NORMAL shall be located on the front panel of the driver amplifier for resetting the overload circuit after the fault is corrected. The switch shall be spring-loaded to return to the NORMAL position.

3.3.4.10.5 Local key/normal switch. A three-position local key/normal switch shall be located on the front panel of the driver amplifier for maintenance purposes. Two switch positions shall be labeled LOCAL KEY. The remaining center position shall be labeled NORMAL. The switch shall be spring-loaded to return to NORMAL when released from one LOCAL KEY position and shall lock in the other LOCAL KEY position. The driver amplifier and the AM-6047B/FRT-84(V) shall be keyed to transmit in either LOCAL KEY position.

3.3.4.11 Driver amplifier RF sampling.

3.3.4.11.1 Rear panel input connector. One coaxial connector of 50-ohms impedance shall be mounted on the rear panel of the driver amplifier and shall be labeled INPUT MON. The connector shall mate with a standard 50-ohm Type BNC fitting and shall be connected internally to monitor the AM-6047B/FRT-84(V) input.

3.3.4.11.2 Front panel input connector. One coaxial connector of 50-ohms impedance shall be mounted on the front panel of the driver amplifier and shall be labeled INPUT MON. The connector shall mate with a standard 50-ohm Type BNC fitting and shall be connected internally to monitor the AM-6047B/FRT-84(V) input.

3.3.5 Interconnecting box front panel indicators. The front panel of the J-2822B/FRT-84(V) shall have a power indicator lamp, illuminated in white to indicate when power is applied, and a manual tune indicator, illuminated in red when manual tune is selected at the AM-6047B/FRT-84(V) and the driver amplifier is manually tuned.

3.3.6 PP-6067A/FRT design requirements. The PP-6067A/FRT shall produce the high voltages for the driver amplifier from a three-phase, 220 VAC, 47 Hz to 63 Hz source. Connections to the PP-6067A/FRT shall be made at the rear of the case.

3.3.6.1 Front panel indicators. A 600 V, 1.5 A indicating fuseholder Type FHL35W in accordance with MIL-F-19207/24, a 24 V, 8A fuse and indicator to protect the 400 Hz inverter circuit from overload, and a power-on indicator shall be located on the PP-6067A/FRT front panel.

3.3.7 AM-6047B/FRT-84(V) design requirements. The AM-6047B/FRT-84(V), hereinafter referred to as the final power amplifier (PA), shall be a single-stage, linear PA that shall use a 4CX15000 tube to produce a nominal output of 10 kW PEP or 5 kW avg power with a nominal input of 800 W PEP.

3.3.7.1 PA primary power. Primary power for the PA and its associated power supply shall be transformed to 220 V by the autotransformer in the bottom of the exciter-driver cabinet.

3.3.7.2 PA power supply. The PA power supply shall provide all the necessary operating voltages for the PA and shall be an integral part of the amplifier. Circuit breakers with resets on the front panel shall protect all circuits from damage. Short circuits applied to any power supply output shall cause no damage to the power supply. Solid-state silicon devices shall be used throughout the power supply.

3.3.7.2.1 Automatic in-rush transient current limiting. For definition of in-rush current, see 6.4.2. Automatic in-rush transient current limiting circuitry shall be provided in each of the three main AC input phases of the PA high-voltage power supply to limit the primary

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power transformer in-rush current to the limits specified herein. The primary purpose of this circuit shall be to preclude circuit breaker actuation and to protect other components in the transmitter from damage caused by extremely high current in the primary windings of the power transformer immediately following transmitter power turn-on. During the initial period of 106.38 ms following the application of AC line voltage to the power transformer, in-rush transient currents shall not exceed a peak value of 50 percent greater than the nominal AC input current value attained when the transmitter is providing 5 kW avg power output into a 50-ohm load with the AC main source set at 200 V, 47 Hz. The main AC line current values present under the conditions specified herein shall be used as the reference currents. At the end of the initial period of 106.38 ms after transmitter power turn-on, the in-rush transient current shall decay to values no greater than 3 percent above the reference currents. This requirement shall apply for all combinations of prime power voltages and frequencies at which the transmitter is required to operate.

3.3.7.2.1.1 Multi-step-start circuit. The use of a multi-step-start circuit to control the limits of power transformer in-rush transient currents is required. The operation of the multi-step-start circuit shall be fully automatic and shall be capable of providing full protection of the transmitter when short term power interruptions occur during the stepping cycle. When primary AC power interruptions which are less than 106.38 ms but greater than 10 ms in duration occur, the step-start circuit shall automatically recycle to step 1 prior to the application of voltage to the power transformer. The recycling process shall also apply to long term power interruptions and normal power shut-off at the power source or by the transmitter controls. This circuit shall be activated and shall function through the stepping sequence in each case where primary power is to be applied to the high-voltage transformer. An additional period of 106.38 ms is permitted for each step function used, but in no case shall the number of steps exceed three.

3.3.7.2.1.1.1 Electrical contactors. The electrical contactors used in the step-start circuit shall provide failure-free operations, when the transmitter is powered by a source of 200 V, 47 Hz with a negligible source impedance, adequate to comply with the MTBF requirement specified in 3.4.1.

3.3.7.2.1.1.2 Step-start circuit resistors. Current limiting resistors shall be of the wire-wound type and shall be rated at 75 A peak, 60 A for 100 ms.

3.3.7.2.2 B-plus filter charge characteristic. The initial B-plus filter charge characteristic shall be chosen such that, when acting in conjunction with the step-start circuit specified in 3.3.7.2.1.1, the transmitter shall not generate transient currents that exceed the limits specified herein.

3.3.7.3 Frequency range and tuning. The frequency range of the PA shall be continuous from 2.0000 MHz to 29.9999 MHz plus or minus half the transmitted intelligence bandwidth. Course tuning information shall be provided to the final PA from the exciter through the band repeater assembly using a five-level binary coded signal and potentials not exceeding 30 VDC.

3.3.7.3.1 Automatic tuning system. The portions of the automatic tuning system concerned with tuning and matching the output stage to the transmitter load impedance shall remain active at the completion of the tuning cycle. Controls shall be provided so that at the completion of a tuning cycle the sensitivity of the tuning system may be reduced at least to the extent that 10 percent changes in the network resonant frequency or plate load impedance shall be corrected. Automatic tuning to restore the desired conditions of tuning and loading shall not interrupt the output signal. All stages of the PA not concerned with output stage loading shall be broadband in segments of the 2.0000 MHz through 29.9999 MHz tuning range. Selection of the desired segment shall require only the coarse tuning information specified in 3.3.7.3.

3.3.7.4 TGC circuit. An electronic TGC circuit shall be provided in the PA to generate a control voltage which, when interfaced with the exciter, shall automatically adjust the transmitter power output during the tuning cycle so that power output during operation is set to rated avg power within a ±1.0 dB limit. Adjustable controls associated with this circuit shall be equipped with locking devices. The TGC system shall be of the infinite memory type.

3.3.7.5 PPC circuit. A PPC circuit shall be provided in the PA to generate a preset control voltage which, when interfaced with the exciter, shall prevent the peak power output of the transmitter from exceeding the limits of this specification. Adjustable level controls associated with the PPC circuit shall be equipped with locking devices and shall be located inside the PA. The circuitry shall be such that special readjustment of the controls is not required when changing operating modes, for example, changing from A3J to A3B.

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3.3.7.6 PA input impedance. Throughout the frequency range of 2.0000 MHz through 29.9999 MHz the PA input circuit impedance shall be nominal 50-ohms resistive with reactive components reduced to values which will cause a VSWR no higher than 1.5:1 at the exciter output.

3.3.7.7 VSWR output monitor. A VSWR monitor circuit shall be provided for automatically reducing the PA power output without interruption of traffic when the reflected power is equivalent to that produced by a VSWR of between 2.5:1 and 4.0:1 at rated output power. The circuit shall activate an alarm and disable the transmitter if the reflected power exceeds that produced by a VSWR of 4.0:1 at rated power output. A remote and local indication of the fault shall be provided on the exciter and the RCU.

3.3.7.8 Final PA assembly. The final PA assembly shall have seven frequency bands with the following ranges of frequency:

a. Two MHz to 4 MHz

b. Four MHz to 8 MHz

c. Eight MHz to 12 MHz

d. Twelve MHz to 16 MHz

e. Sixteen MHz to 20 MHz

f. Twenty MHz to 24 MHz

g. Twenty-four MHz to 30 MHz

3.3.7.8.1 Frequency band switching. Switching shall be accomplished by the five-wire code from the band repeater assembly to the RF band switch in the 2 MHz to 4 MHz, 4 MHz to 8 MHz, 8 MHz to 12 MHz, 12 MHz to 16 MHz, 16 MHz to 20 MHz, 20 MHz to 24 MHz and 24 MHz to 30 MHz bands. When the 12 MHz to 16 MHz and 16 MHz to 20 MHz or the 20 MHz to 24 MHz and the 24 MHz to 30 MHz bands are selected on the switch, further selection of the 12 MHz to 16 MHz or 16 MHz to 20 MHz band and the 20 MHz to 24 MHz or 24 MHz to 30 MHz band shall be accomplished by a band relay assembly.

3.3.7.9 RF sampling. Individual RF output sampling receptacles shall be provided to enable the monitoring of RF input (see 3.3.4.11) and output spectra of the transmitter PA and its intermediate stages as specified herein. Sampling shall be provided at a level corresponding to the level produced at the receptacle in the GFE with modulation by a standard two-tone signal and the transmitter delivering full rated 10 kW PEP output into a 50-ohm load, except as specified in 3.3.7.9.1 and 3.3.7.9.2. The intent of the sampling receptacles is to allow monitoring of the transmitted signal within the normally transmitted bandwidth of 12.6 kHz.

3.3.7.9.1 Rear panel output connector. One coaxial connector of 50-ohms impedance shall be mounted on the floor of the final power amplifier cabinet and shall be labeled OUTPUT MON. The connector used for this output shall mate with a standard 50-ohm Type BNC fitting and shall be connected internally to monitor the PA output. The sample level shall be based on 5 kW avg power output.

3.3.7.9.2 Front panel output connector. One coaxial connector of 50-ohms impedance shall be accessible at the front of the PA and shall be labeled OUTPUT MON. The connector shall mate with a standard 50-ohm Type BNC fitting and shall be connected internally to monitor the PA output. The sample level shall be based on 5 kW avg power output.

3.3.7.10 PA standby. The PA standby condition is identical to the exciter standby condition specified in 6.4.5 with the following addition: All components and circuitry in the PA requiring more than 1 s to reach full operating condition shall remain activated during standby. All circuitry remaining active during the standby condition shall be protected so that the cabinet may be opened for maintenance without violating personnel safety requirements (see 3.3.1.19).

3.3.7.11 Tuning capacitor 9A1C8. Tuning capacitor 9A1C8, a variable, servo-controlled capacitor, shall be vacuum ceramic or glass type.

3.3.7.12 RF loading capacitors 9A1C4 and 9A1C6. RF loading capacitors 9A1C4 and 9A1C6 (variable, servo-controlled, parallel connected capacitors) shall be vacuum ceramic or glass type.

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3.3.7.13 DC isolation capacitors. To preclude antenna damage or hazards to personnel, all capacitors used for RF output coupling (DC blocking from the output line) at the PA plate shall withstand a peak stress equal to 20,000 VDC at the PA plate at an ambient temperature of 50°C, 95 percent relative humidity (RH). The determination of the ability of capacitors to conform to this requirement shall be made before they are incorporated into the transmitter.

3.3.7.14 Capacitor 9A1C10. Capacitor 9A1C10 in the high-voltage (8 kilo-volt (kV) DC) B-plus circuit in the pressurized compartment of the PA shall withstand a minimum of 25 kV DC in a 95 percent RH environment at 50°C ambient temperature at 30 MHz without arcing or breakdown. A water-tight polytetrafluoroethylene shroud shall be used on the capacitor.

3.3.7.15 RF coupling capacitors. To preclude personnel injury or antenna damage, all capacitors used for RF coupling in the PA shall withstand a peak stress voltage equal to 20,000 VDC at the PA plate at 30 MHz.

3.3.7.16 Tank inductors and RF coils. All metal parts in the PA RF coils, except hardware and end brackets, shall be silver plated in accordance with QQ-S-365, Type II, Grade A, to a thickness of 0.0002 in. to 0.0005 in. directly over the copper. A nickel undercoat shall not be used.

3.3.7.17 RF switches. PA RF switches shall be 10-position rotary power selector switches with two decks paralleled back to back, with six active positions, continuously rotating without detents or steps. Contact shoes shall be coin silver. Current carrying members shall be deburred to a minimum radius of 0.254 cm (0.10 in.) and shall be silver plated to a minimum thickness of 0.0005 in. RF switches shall be forced air-cooled by air directed from the pressurized PA compartment.

3.3.7.18 Plate supply rectifier. The PA plate supply rectifier shall be a silicon high-voltage potted assembly with minimum ratings of 10,000 V peak reverse voltage and 7000 V rms input, a maximum leakage current of 0.3 mA at 100°C, and an ambient operating temperature range of -65°C to +120°C.

3.3.7.19 PA meters. PA meters shall be located on the final power amplifier cabinet meter panel at the top front of the cabinet and visible when the cabinet doors are closed. The DC amplifier assembly (A1A5) and the monitor/control assembly (A1A6), shall be located behind the cabinet door. Two meters associated with PA parameters are located on the front panel of the J-2822B/FRT-84(V) (see 3.3.7.19.1 and 3.3.7.19.2).

3.3.7.19.1 Elapsed time meter. An elapsed time meter shall be provided in the transmitter to indicate operating hours, defined as high-voltage ON time. A meter to record filament hours for the PA electron tube shall be provided on the J-2822B/FRT-84(V) front panel. The elapsed time meter shall be calibrated for operation on 60-Hz power.

3.3.7.19.2 Tune cycle counter. A digital recording counter shall be provided in the transmitter to record the number of tune cycles accomplished. This feature shall apply to the automatic and manual tuning modes. The counter shall be capable of recording and displaying in an unambiguous manner up to 9999 events and shall be capable of being reset to 0. The reset feature shall be suitably protected so that the counter cannot be inadvertently reset. The counter shall be mounted on the front panel of the J-2822B/FRT-84(V). The tune cycle counter shall be calibrated for operation on 60-Hz power.

3.3.7.19.3 RF output power meter. The RF output power meter shall be located on the cabinet meter panel and shall provide scales and switching for measurement of forward and reflected power as follows:

Forward	Reflected
0 kW to 15 kW PEP	0 kW to 15 kW PEP
0 kW to 15 kW avg	0 kW to 15 kW avg
	0 kW to 1.5 kW PEP
	0 kW to 1.5 kW avg

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The scales shall be calibrated logarithmically with the minimum scale calibration mark of 1 kW for the 15 kW scale and 0.1 kW for the 1.5 kW scale. One switch shall be of the two-position rotary type and shall select either PEP or avg power indications. A spring-loaded, three-position switch shall select the forward and reflected power indications and shall be arranged so that forward power is indicated in the center position. The spring-loading shall provide automatic return of the switch to the forward power (center) position. Calibration of the RF output power meter shall be accurate to within 10 percent over the 2.0000 MHz through 29.9999 MHz frequency range for single-tone, two-tone, 16-tone conventional tone library used with teletype terminal equipment, and noise-loaded signals. A panel with provisions for screwdriver adjustment of the meter calibration shall be located inside the final power amplifier cabinet front doors just below the meter panel.

3.3.7.19.4 Plate current meter. The PA plate current meter, located on the cabinet meter panel, shall indicate from 0 A to 5 A.

3.3.7.19.5 PA multimeter and switch. The PA multimeter and an associated 15-position switch shall be located on the front of the DC amplifier assembly (A1A5) as in the GFE. A table displaying switch functions as specified in TABLE I, associated meter indications and other pertinent information shall provide data identical to that provided by the GFE and shall be located as in the GFE. The DC amplifier assembly panel shall incorporate a pre-position chart that displays the pre-position meter indications for the tune and load variable capacitors.

TABLE I. PA multimeter switch functions.

Multimeter switch			
Switch position	Label	Function	Miscellaneous
1	VK	RF input at cathode	Normal is same meter deflection as position 2
2	VP	RF plate V	6 kV full scale
3	IS	Avg screen current	60 mA full scale
4	PF	Avg forward power	10 kW approx +3 on scale
5	PR	Avg reflected power	3 kW approx +1.5 on scale
6	TUNE SET	Tune cap band pre-set	Null when pre-positioned
7	TUNE POS	Tune cap position analog	Minus = CCW, plus = CW
8	TUNE DET	Tune detector error V	Null when fine tuned
9	LOAD DET	Load detector error V	Null when fine tuned
10	LOAD POS	Load cap position analog	Minus = CCW, plus = CW
11	LOAD SET	Load cap band pre-set	Null when pre-positioned
12	6 VDC	Use with external probe	770 K input impedance
13	60 VDC	Use with external probe	1.10 M input impedance
14	600 VDC	Use with external probe	1.10 M input impedance
15	ZERO	Meter check	Mechanical zero check

3.3.7.19.6 PA fault detector meter and switch. A PA fault detector meter and associated 18-position switch shall be located on the monitor control assembly (A1A6) as in the GFE. A table displaying switch functions as specified in TABLE II, associated meter indications and other pertinent information shall provide data identical to that provided by the GFE and shall be located as in the GFE. A FAULT RESET pushbutton on the assembly shall reset the PA fault detectors when pressed.

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TABLE II. PA fault detector switch functions.

Fault detector switch			
Switch position	Label	Function	Miscellaneous
1	LINE	Primary power input voltage	From auto transformer
2	28 V	Control circuit	Unregulated
3	FIL	Filament voltage	At primary of transformer
4	+15	Meter 8 servo A	Regulated
5	-15	Meter 8 servo A	Regulated
6	BIAS	Grid bias voltage	Over 0.5 unkeyed 0.25 keyed
7	+1 kV	Screen voltage	Zero after fault indicated
8	+8 kV	Plate voltage	Zero after fault indicated
9	IS OVERLOAD	Excessive screen current	Reset button (100 mA)
10	IP OVERLOAD	Excessive plate current	Reset button (3.5 A)
11	SWR OVERLOAD	Excessive reflected power	Reset button (3 kW)
12	TUNE TIME	Excess tune time	Reset button (15 s)
13	TUNE LIMIT	Capacitor limit switch	Turn capacitors manually
14	STICK INTLK	High-voltage shorting stick	Replace stick
15	SLIDE INTLK	PA slide assembly	Close slide
16	CB INTLK	Cabinet bottom access	Shut access cover
17	PL INTLK	Plate compartment	Replace cover
18	IN INTLK	Input compartment	Replace cover

3.3.7.20 PA controls and indicators.

3.3.7.20.1 Manual tuning controls. Manual tuning controls, located on the monitor control assembly (A1A6), shall function and be labeled as in the GFE. Manual tuning shall be accomplished by selecting manual tune, depressing the tune power request switch to remove RF power from the tuning circuits, and depressing the band advance control to step the final transformer assembly one band at a time until the desired band appears in the window on the band repeater assembly. Manual adjustment of the PA tune and load capacitors to minimum plate current and maximum power output shall be accomplished as in the GFE to complete the manual tuning process.

3.3.7.20.2 Tune detector assembly (9A1A3) controls. Controls located on tune detector assembly (9A1A3) shall permit trim adjustment of the tune discriminator and the load detector capacitors.

3.3.7.20.3 Final power amplifier cabinet meter panel indicators. Cabinet meter panel indicators shall be provided as follows:

a. Fault indicator: The fault indicator shall be illuminated in red to provide a visual indication of a fault in the PA.

b. Manual tune indicator: The manual tune indicator shall be illuminated in amber to indicate selection of PA manual tuning.

3.3.7.20.4 Tune up control indicators. Tune up control indicators consisting of a preposition lamp (white), a tune power lamp (blue), an RF at plate lamp (green), and a tuning lamp (red) shall function as in the GFE. These indicators shall be located on the monitor/control assembly (A1A6) as in the GFE.

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3.3.7.21 PA circuit breaker panel. The PA circuit breaker panel shall provide mounting for circuit breakers for the final power amplifier cabinet blower power circuit, PA filament, plate and screen circuits, and the 115 V and 28 V power circuits. Circuit breakers shall comply with 3.3.1.2.8.

3.3.7.22 Band repeater assembly. The band repeater assembly shall, in response to a five-wire code generated in the exciter when a new operating frequency is selected, reposition and generate band codes. Band codes from the band repeater assembly shall be applied to the driver amplifier (see 3.3.4.5) to cause band selection in the pretuned final transformer assembly and to the PA (see 3.3.7.8.1) to pre-position the band switch circuits to coarse tune the PA. The selected frequency range, corresponding to the range of the selected PA final transformer assembly band, shall be displayed in a window in the repeater assembly. The band repeater assembly shall be located in the final power amplifier cabinet.

3.3.8 Interchangeability. Like units, modules, PCBs, assemblies, sub-assemblies, and replaceable parts shall be physically, electrically, and functionally interchangeable with corresponding items in the AN/FRT-83(V), AN/FRT-83A(V), AN/FRT-83B(V), AN/FRT-84(V), and AN/FRT-84A(V). Individual items shall not be hand-picked for fit or performance. Interchangeability requirements shall be two-way.

3.3.8.1 Exciter interchangeability. The exciter produced to the requirements of this specification shall be interchangeable with the exciter in the AN/FRT-83(V), AN/FRT-83A(V), AN/ FRT-83B(V), AN/FRT-84(V), AN/FRT-84A(V), and the AN/FRT-85(V) specified in MIL-T-28784.

3.3.8.2 Driver amplifier and PP-6067A/FRT interchangeability. The driver amplifier and the PP-6067A/FRT produced to the requirements of this specification shall be interchangeable with the AM-6046B/FRT and the PP-6067A/FRT in the AN/FRT-83(V), AN/FRT-83A(V), AN/FRT-83B(V), AN/FRT-84(V), and the AN/FRT-84A(V).

3.3.8.3 Final power amplifier and associated power supply interchangeability. The final power amplifier and its associated power supply shall be interchangeable with like units in the AN/FRT-84(V) and the AN/FRT-84A(V).

3.3.8.4 Electronic keyer interchangeability. The electronic keyer FSK produced to the requirements of this specification, shall be interchangeable with the KY-655/FRT incorporated in the AN/FRT-83(V), AN/FRT-83A(V), AN/FRT-83B(V), AN/FRT-84(V), AN/FRT-84A(V), and the AN/FRT-85(V) specified in MIL-T-28784.

3.3.8.5 Remote control system LCU interchangeability. The transmitter shall be designed to permit installation of the LCU of the AN/FSQ-98(V) and AN/FSQ-98A(V) and operation under the control of the RCU of the AN/FSQ-98(V) and AN/FSQ-98A(V). No engineering changes or modifications to the transmitter shall be necessary to conform to this requirement. LCU installation shall be accomplished by making simple cable connections without soldering or the use of cable connector adaptors. The transmitter shall be equipped with a blank panel which can be mounted in the exciter-driver cabinet equipment rack to cover the void that exists when the LCU is not installed.

3.3.9 Remote control system. The transmitter shall be capable of being remotely controlled by the AN/FSQ-98(V) and AN/FSQ-98A(V) RCS or an equivalent system which conforms to the requirements specified herein and which interfaces with the transmitter through the exciter. The extent of the remote control shall be limited to those functions listed in TABLES III, IV, and V.

TABLE III. Exciter connector J7 MS3116E20-39P.

Terminal	Function	Remarks
A		
B		
C	100 Hz	
D		
E		
F	1 kHz	
G		
H		
J		

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TABLE III. Exciter connector J7 MS3116E20-39P (Cont).

Terminal	Function	Remarks
K	10 kHz	Frequency selection (see TABLE V)
L		
M		
N		
P	100 kHz	
R		
S		
T		
U	1 MHz	
V		
W		
X	10 MHz	
Y		
Z	VCO A	
a	VCO B	
b	AO	
c	A2, A3H	Class of emission selection
d	A1, F, F4	
e	SSB -10 dB carrier	
f	SSB -20 dB carrier	
g	SSB -40 dB carrier	
h	SSB -infinity carrier	
i	Standby command	
j	Operate command	
k	Ground	
m	Ground	
n	PTT command	

TABLE IV. Exciter connector J8 MS3116E20-39PW.

Terminal	Function
A	USB select control
B	LSB select control
C	ISB (2) select control
D	ISB (4) select control
G	AO class of emission readback
H	A2, A3H class of emission readback
J	A1, F1, F4 class of emission readback
K	SSB -10 class of emission readback
L	SSB -20 class of emission readback
M	SSB -40 class of emission readback
N	SSB -infinity class of emission readback
P	Operate readback
R	Standby readback
S	USB readback
T	LSB readback
U	ISB (2) readback
V	ISB (4) readback
Y	Ready readback (common)
Z	Ready readback
a	Fault readback
b	Fault readback (common)
c	Local status
d	Transmitter not available readback (common)
e	Tune command
f	Tune activate common (ground)

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TABLE IV. Exciter connector J8 MS3116E20-39PW (Cont).

Terminal	Function
g	+5 VDC
h	PTT readback
i	Transmitter fail
j	VFG readback
k	Standby-operate readback (common)
m	Sideband select (common)
n	Sideband readback (common)
q	Class of emission readback (common)
v	VFG command

TABLE V. Frequency selection tables.

100 Hz digit selection				
Digit	J7-A	J7-B	J7-C	J7-D
0	0	0	0	1
1	1	0	0	1
2	0	0	1	1
3	1	0	1	1
4	0	1	1	0
5	1	1	1	0
6	0	1	0	0
7	1	1	0	0
8	0	0	0	0
9	1	0	0	0
1 kHz digit selection				
Digit	J7-E	J7-F	J7-G	J7-H
0	1	0	0	1
1	0	0	0	1
2	1	0	1	1
3	0	0	1	1
4	1	1	1	0
5	0	1	1	0
6	1	1	0	0
7	0	1	0	0
8	1	0	0	0
9	0	0	0	0
10 kHz digit selection				
Digit	J7-J	J7-K	J7-L	J7-M
0	0	0	1	1
1	1	1	0	1
2	0	1	0	1
3	1	0	0	1

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TABLE V. Frequency selection tables (Cont).

4	0	0	0	1
5	1	1	1	0
6	0	1	1	0
7	1	0	1	0
8	0	0	1	0
9	1	1	0	0

100 kHz digit selection				
Digit	J7-N	J7-P	J7-R	J7-S
0	0	0	0	1
1	1	0	0	1
2	0	0	1	1
3	1	0	1	1
4	0	1	1	0
5	1	1	1	0
6	0	1	0	0
7	1	1	0	0
8	0	1	0	0
9	1	0	0	0

1 MHz digit selection				
Digit	J7-T	J7-U	J7-V	J7-W
0	1	1	0	0
1	0	1	0	0
2	1	0	0	0
3	0	0	0	1
4	1	0	0	1
5	0	0	0	1
6	1	0	1	1
7	0	0	1	1
8	1	1	1	1
9	0	1	1	0

10 MHz digit selection		
Digit	J7-X	J7-Y
0	1	1
1	0	1
2	1	0

VCO selection		
Frequency range (MHz)	J7-Z (VCO A)	J7-a (VCO B)
2.0000 to 11.9999	0	0
12.0000 to 21.9999	0	1
22.0000 to 29.9999	1	0

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3.3.9.1 Remote control interface. All interfaces between the RCS and the transmitter shall be via the remote system LCU. All control signals necessary to satisfy system operation shall be exchanged between the LCU and the exciter.

3.3.9.2 LCU location. The exciter-driver cabinet shall provide a space capable of accepting the LCU. The LCU shall be contained in an enclosed case suitable for mounting in a standard 48.3 cm (19 in.) equipment rack as specified in MIL-STD-189. Its dimensions shall not exceed 18 cm (7 in.) high, 45.7 cm (18 in.) deep, and 43.5 cm (17.125 in.) wide, exclusive of the front panel mounting. The LCU shall be located immediately below the keyer and the exciter in the exciter-driver cabinet. Primary power for the LCU shall be available on the standard multiple outlet strip specified in 3.3.1.2.7.1.

3.3.9.3 Transmitter control. All information received by the LCU from the RCU shall be converted to parallel data for application to the exciter control lines. The remote-controlled signalling shall be decoded as necessary in the LCU to achieve compatibility with the exciter control lines and functions. Monitor information describing the transmitter status shall be encoded by the LCU for transmission to the RCU.

3.3.9.3.1 Transmitter control and monitor connections. Actual control of the transmitter and monitoring of its state of operation when in a remote control mode shall be via two connectors located on the exciter rear panel. Cables shall be provided with the LCU to mate with these connectors. The connector types and pin functions are specified in TABLES III, IV, and V.

3.4 Reliability requirements.

3.4.1 Quantitative reliability requirements. The upper test MTBF of the transmitter (θ_0 as defined in MIL-STD-781) shall be 600 hours. The lower test MBTF (θ_1 as defined in MIL-STD-781) shall be 300 hours.

3.5 Construction.

3.5.1 Modular construction. The use of modular construction shall be in accordance with MIL-E-16400 except that the standard electronic module (SEM) requirements of MIL-STD-1378 shall not apply. All modules which make up the major components of the exciter, the driver amplifier, and the PA shall be completely removable by a single disconnect, except for the following:

a. Driver amplifier:
    1. VSWR bridge
    2. Final tank assembly
    3. Driver tank assembly
    4. Driver tube assembly
b. PA:
    1. Tune detector (9A1) assembly
    2. High-voltage rectifier assembly
    3. Step-start assembly
    4. Band relay assembly
    5. Fault control assembly
    6. 28 V power supply

3.5.2 Cabinet construction. The transmitter cabinets and their internal panels and plates shall be constructed of material of sufficient strength to provide a rugged and durable enclosure adequate for either fixed-station or transportable service. Cabinet frames, side panels, and doors shall be constructed of steel. Internal panels and plates shall be constructed of aluminum. Construction of the cabinets shall be consistent with the shock requirement specified in 3.8.2. Openings which would otherwise admit rodents, insects, or dirt shall be screened

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or equipped with replaceable air filters. A wiring port for external and interconnecting power, signal and control wiring shall be located in the base of each cabinet. A solid cover plate shall be installed over each wiring port prior to offering the transmitter for delivery.

3.5.2.1 Side panel removal. The cabinets shall be assembled with captive nuts or nut bars to permit removal of damaged side panels without the necessity for total disassembly (or replacement) of an individual cabinet.

3.5.2.2 Cabinet doors and fasteners. The final power amplifier cabinet shall be equipped with front hinged doors. The doors shall cover the entire front of the final power amplifier cabinet except for the meter panel and the circuit breaker panel. The exciter-driver cabinet shall be equipped with front and rear hinged doors. The exciter-driver cabinet front door shall cover the driver amplifier front panel, the PP-6067A/FRT front panel, and the protective screen. Door hinges shall be either full-length and continuous or, as a minimum, positioned at three points. Door fasteners (latches) shall be of a quarter-turn handle design.

3.5.2.3 Support strength. Drawer slides, detents, mounting surfaces, and other attachment mechanisms which support equipment shall be designed to support twice the weight of the equipment supported.

3.5.2.4 Tip stability. Design of rack-mounted equipment shall maintain the center of gravity as low as possible.

3.5.2.5 Drawer slide security. Equipment design shall include positive mechanical means to prevent accidental derailing and extension of equipment mounted on drawer slides.

3.5.3 Degree of enclosure. The degree of enclosure shall be protected, conforming to the definition in 6.4.6.

3.5.4 Mountings. All major units shall be mounted on slides, tracks, or hinges to permit servicing of individual units from the front of the transmitter. For all units weighing less than 23 kg (50.6 lb), the mounting shall also provide for a +90 degree and a -90 degree front-to-back tilting to a horizontal position in steps of 45 degrees with provisions for locking (hold) at each 45 degree step. The -90 degree tilting requirement shall not apply to the bottom slide or track-mounted units. Major units not mounted on slides or tracks shall be readily accessible from the front of the transmitter for servicing purposes.

3.5.4.1 Transmitter mounting. The exciter-driver cabinet and the final power amplifier cabinet shall each be drilled for the use of floor mounting bolts when the transmitter is installed.

3.5.4.2 Exciter mounting. The exciter chassis and panel assembly shall be slide-mounted to permit extending the chassis from its own case on slides and rotating the assembly to either of two vertical positions (±90 degrees), in steps of 45 degrees, for servicing.

3.5.4.3 Electronic keyer mounting. The electronic keyer chassis and panel assembly shall be slide-mounted to permit extending the chassis from its own case and rotating the assembly to eitner of two positions (±45 degrees from the horizontal) for servicing.

3.5.4.4 J-2822B/FRT-84(V) mounting. The J-2822B/FRT-84(V) chassis and panel assembly shall be slide-mounted to allow the assembly to be extended and rotated to either of two vertical positions (±90 degrees), in steps of 45 degrees, for servicing.

3.5.4.5 Driver amplifier mounting. Driver amplifier assemblies shall be secured to a chassis and panel assembly that is slide-mounted to allow it to be extended and rotated to a vertical position for servicing.

3.5.4.6 PP-6067A/FRT mounting. PP-6067A/FRT components shall be mounted on a chassis and panel assembly that shall be hinge-mounted in a metal case. The chassis and panel assembly shall be constructed to permit dropping it to a horizontal position for servicing. Power transformers for the PP-6067A/FRT shall be constructed as an integral part of the case.

3.5.4.7 PA mounting. The PA chassis assembly shall be mounted on slides that allow it to be extended from the front of the final power amplifier cabinet for servicing. PA assemblies not mounted on the PA chassis shall be mounted on the interior of the final power amplifier cabinet.

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3.5.5 Retractable cable harness. Cable harnesses associated with slide-mounted assemblies shall be equipped with a flat-pack-type (molded) cable harness and retractable carrier assembly as in the GFE. The carrier assembly shall be constructed to preclude damage to the cable harness during all conditions of assembly, removal, insertion, or tilting, at any angle specified herein.

3.6 Maintainability requirements.

3.6.1 Quantitative maintainability requirements. The transmitter shall have a corrective-maintenance-time (M_{ct}) of 1.0 hour and a maximum-corrective-maintenance time (M_{maxct}) of 1.5 hours at a 90 percent confidence level when repair is accomplished at the organizational level of maintenance by the replacement of modules and chassis-mounted components (this includes electronic, electrical, electromechanical, and mechanical parts). The M_{ct} includes localization, disassembly, interchange, reassembly, alignment, and checkout for all maintenance tasks. For purposes of maintainability, a module is considered the lowest replaceable subassembly (PCB) for which repair is accomplished by the replacement of piece parts.

3.7 Performance characteristics.

3.7.1 General performance characteristics.

3.7.1.1 Frequency range. The carrier frequency range of the transmitter shall be from 2.0000 MHz through 29.9999 MHz (plus and minus the intelligence side-bands) in steps of 100 Hz.

3.7.1.1.1 Tuning time. The transmitter shall be capable of being automatically tuned and loaded on any operating frequency from 2.0000 MHz to 29.9999 MHz in 15 s or less when the output is terminated in a load producing a VSWR of up to 4:1.

3.7.1.2 Operating modes. The transmitter shall provide the following modes of operation or types of emission:

a. AØ: RF carrier present

b. A1: CW telegraphy, interrupted RF carrier

c. A2: Modulated CW telegraphy

d. A3E: Compatible amplitude modulated telephony (AME simplex) (USB)

e. A3A: SSB simplex, reduced carrier (USB or LSB)

f. A3B: Two ISB (NORMAL and SIMPLEX)

g. A3J: SSB simplex, suppressed carrier (USB and LSB normal), (SSB simplex PTT (USB))

h. A9B: Two ISB with combined telephony and telegraphy

i. A9J: Four ISB with combined telephony and telegraphy

j. F1: FSK (USB)

k. F4: FAX (USB)

3.7.1.3 Power output(s) for specified input(s). Exciter, driver amplifier, and PA power output(s) for specified input(s) shall be in accordance with TABLE VI.

TABLE VI. Power output(s) for specified input(s).

Mode	Exciter output and driver amplifier input	Driver amplifier output and final power amplifier input	Final power amplifier output
A1	100 mW avg	400 W avg	5 kW avg
A2	100 mW avg	400 W avg	5 kW avg

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TABLE VI. Power output(s) for specified input(s) (Cont).

Mode	Exciter output and driver amplifier input	Driver amplifier output and final power amplifier input	Final power amplifier output
A3E	200 mW PEP (two-tone)	800 W PEP (two-tone)	10 kW PEP (two-tone)
A3A	200 mW PEP (two-tone)	800 W PEP	10 kW PEP
A3B	200 mW PEP (two-tones in each sideband)	800 W PEP	10 kW PEP
A3J	200 mW PEP (two-tone)	800 W PEP (two-tone)	10 kW PEP (two-tone)
A9	200 mW PEP (two-tones in each sideband)	800 W PEP	10 kW PEP
F1	100 mW avg	400 W avg	5 kW avg
F4	100 mW avg	400 W avg	5 kW avg

3.7.1.4 Electromagnetic field. The transmitter shall operate without derating or degradation of performance when immersed in an RF field of 10 kHz through 30 MHz at a power density level of 0.01 W/cm^2. See 3.3.1.20 for protection of tuning and conditioning circuits.

3.7.1.5 In-band electrical noise. For definition of in-band electrical noise, see 6.4.1 The in-band electrical noise of the transmitter shall be at least 60 dB below the transmitter full rated 5 kW avg power output.

3.7.1.5.1 Key-up. The transmitter shall be capable of employment in simplex operation without causing electrical interference while the transmitter is in the key-up or receive-mode condition. When the transmitter is in the key-up or receive-mode condition, the transmitter avg noise power output measured in a 3000-Hz bandwidth anywhere between 2.0000 MHz and 29.9999 MHz shall be at least 85 dB below 1 mW.

3.7.1.5.2 Intelligence loss. When the transmitter is changed from key-up or receive-mode condition to key-down (transmit) condition, not more than 10 ms of the intelligence in the modulating signal shall be lost.

3.7.1.6 Envelope delay. The overall envelope delay distortion for each sideband shall not exceed the following:

Delay	Frequency
500 microseconds (μs)	382 Hz to 3020 Hz
1000 μs	370 Hz to 382 Hz
1000 μs	3020 Hz to 3040 Hz

3.7.2 Exciter performance characteristics.

3.7.2.1 Frequency stability. The frequency stability of the entire transmitter, determined by the exciter 1-MHz internal standard, shall be no worse than 1 part in 10^8 when measured at the RF power output point during any 24-hour period at any frequency covered by this transmitter. The absolute permissible frequency error shall not exceed 0.3 Hz in any 24-hour period. The rms phase deviation shall not exceed three degrees in 10 ms.

3.7.2.1.1 Internal frequency standard error and aging.

3.7.2.1.1.1 Absolute permissible frequency error. For definition of absolute permissible frequency error, see 6.4.3.

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3.7.2.1.1.1.1 Aging. The absolute frequency error shall not change more than 0.3 Hz daily, except during the initial aging period. (The initial aging period shall have been completed prior to the time that the transmitter is offered for Government tests). The long-term absolute frequency error shall not exceed the following limits:

Time since last frequency calibration	Allowable absolute error at 29.9999 MHz
Less than 24 hours	0.3 Hz
Between 1 day and 7 days	0.3 Hz X days or 1.5 Hz, whichever is less
Between 7 days and 90 days	1.5 Hz X weeks or 3.0 Hz, whichever is less

3.7.2.2 Interference. Procurement of equipments and subsystems paragraph of MIL-STD-461, having met other electromagnetic interference (EMI) requirements, is applicable to this specification. The radiated and conducted interference produced by the exciter when operated into a 50-ohm resistive load shall not exceed the limits specified in MIL-STD-461.

3.7.2.3 Signal inputs. The exciter shall accept CW, FSK, and FAX audio modulating signals from the electronic keyer and modulation from nominal 600-ohm audio lines. An appropriately labeled channel test jack field shall be installed internally in the exciter to permit the insertion of audio test tones into each of the four audio channels.

3.7.2.3.1 Modulating signals (audio). The exciter shall be capable of being modulated with voice or composite audio frequency (AF) signals at a nominal level of 0 dBm. Input level attenuators shall be provided for accommodation of input levels from +10 dBm to -30 dBm in steps of not more than 1 dB.

3.7.2.3.1.1 Two ISB simplex operation. When SIMPLEX 2 ISB is selected, the exciter shall be capable of being keyed by signals of +6 VDC, with tolerances of +1 V or -0.25 V, applied to the data keyline input. The data keyline connection shall present a minimum load of 5000 ohms to the data keyline. When all of the keyline inputs are between the limits of +0.75 V and 0 V, the exciter shall be in a key-up condition.

3.7.2.3.1.1.1 Keying attack time. The transmitter RF output shall be within 1 dB of steady-state, without ringing, within 8 ms following data keyline closure (audio continuously applied).

3.7.2.3.1.1.2 Keying decay time. The transmitter RF output shall comply with the key-up condition within 8 ms following keyline inputs between +0.75 V and 0 V.

3.7.2.3.1.2. Audio impedance. The input impedance for each of the four voice frequency inputs shall be 600-ohms nominal. There shall be a minimum of 30 dB return loss from 370 Hz to 3040 Hz for each channel.

3.7.2.3.1.2.1 Electrical balance. The electrical balance shall be such that longitudinal currents are at least 40 dB below the reference level (0 dBm at 1000 Hz).

3.7.2.4 Carrier suppression. The carrier shall be suppressed at least 60 dB below the avg power of a single-tone signal at full rated 5 kW output. The class of emission control specifed in 3.3.2.19.2.2 shall accomplish insertion of carrier at levels of 0 dB (no suppression), -6 dB, -10 dB, -20 dB, -40 dB, and -60 dB.

3.7.2.5 Carrier compression. When set for 20 dB carrier suppression, the carrier level shall not be compressed more than 1 dB at any power level from zero to full rated power output.

3.7.2.6 Carrier change under modulation. When the carrier suppression is at maximum (60 dB or more), the carrier level shall not increase more than 2.5 dB at any output level between zero and full rated avg output power.

3.7.2.7 Equal sideband power. When operating in an ISB mode with equal power input to each channel, the total power output shall be divided equally among the channels within ±5 percent.

3.7.2.8 Exciter output impedance. The nominal load impedance of the exciter shall be 50-ohms resistive. Maximum VSWR at the exciter output fitting shall not exceed 1.5:1 throughout

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the required frequency range specified in 3.7.1.1. The exciter output shall be through a connector which mates with a standard 50-ohm Type N fitting.

3.7.2.9 Standby power. Power drain of the exciter in the standby condition shall not exceed 100 W.

3.7.2.10 Transfer time and controls. It shall be possible to place the exciter in standby from full operating condition, or return it to full operating condition from standby, within 1 s, using not more than one front panel control.

3.7.2.11 Exciter response characteristics.

3.7.2.11.1 Four channel ISB performance. When the exciter is configured for four-channel ISB operation, the four channels of the exciter shall be identified in accordance with FIGURE 1.

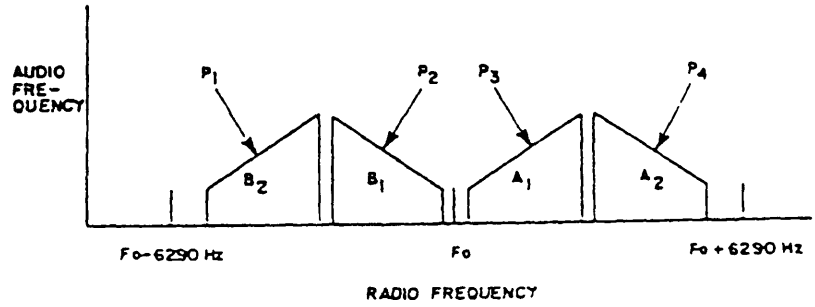


FIGURE 1. Transmitter AF channels.

3.7.2.11.1.1 Output channel inversion. The inversion of channels A_2 and B_2 with respect to channels A_1 and B_1 shall be accomplished by SSB modulation of subcarrier frequencies which are removed 6290 Hz above and below the output carrier frequency.

3.7.2.11.1.2 Channel frequency stability. The frequency stability, including phase jitter, of channels A_2 and B_2 shall be identical with that of channels A_1 and B_1.

3.7.2.11.1.3 Inversion carriers and spurious suppression. The subcarriers used to accomplish the spectra inversion of channels A_2 and B_2 shall not degrade the specified performance requirements of the transmitter. To this end, the carriers and any of the unwanted products of such processes shall be included in the determination of intermodulation products and spurious response of the transmitter.

3.7.2.11.2 In-band response. The response of each audio channel shall be such that the output of any single modulation frequency does not vary, for a constant input level, more than 1 dB (peak-to-valley) over the range of 370 Hz to 3040 Hz for channels A_1, A_2, B_1, and B_2.

3.7.2.11.3 Input and output references. The reference point for each exciter input audio response characteristic shall be 0 dBm, 1000 Hz, sinewave at the audio input terminals of each exciter channel. The audio response of each channel shall be determined by measurement in its respective portion of the modulated RF output of the exciter.

3.7.2.11.3.1 Stability of characteristics. The response of each audio channel of the exciter shall not exceed the limits specified in 3.7.2.11.2 under any allowable conditions of modulation. This requirement means that under any combination of allowable input types and levels in the other three channels, each of the channel response curves shall not exceed the limits specified in 3.7.2.7 and 3.7.2.11.2.

3.7.2.11.4 Out-of-band response. IF filters shall be provided in the exciter with characteristics that will provide the out-of-band performance of the exciter for each channel as specified in TABLE VII. The slope of the out-of-band response curves shall be monotonic. The specified performance shall be met with the exciter installed in the transmitter and operation at any temperature within the ambient range 0°C to 50°C.

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TABLE VII. SB filter characteristics.

Frequency		Attenuation
Channels B1 and A2	Channels A1 and B2	
F_o	F_o	At least -20 dB
F_o +250 Hz	F_o -250 Hz	At least -50 dB
F_o +350 Hz	F_o -350 Hz	At least -60 dB
F_o -3350 Hz	F_o +3350 Hz	At least -20 dB
F_o -3590 Hz	F_o +3590 Hz	At least -60 dB
F_o - Frequency of carrier or subcarrier for each channel		

3.7.2.11.4.1 Sideband filter characteristics. Each sideband filter shall have the following electrical characteristics and shall conform to the interchangeability requirement specified in 3.3.8:

a. Impedance: 50 ohms ±5 percent, in parallel with 0 picofarad (pF) to 25 pF

b. Insertion loss: Not greater than 3 dB ±0.5 dB

c. Input level: Filters shall operate with no degradation of performance with an input level of 250 millivolts (mV) rms, single tone.

d. Spurious return: Spurious return below a frequency of 1.80 MHz shall not be above -60 dB

e. Intermodulation distortion: Filters shall not degrade exciter intermodulation distortion performance of -60 dB referred to either tone of a two-tone test signal within the passband at an input level of 50 mV rms per tone.

f. Reliability: Filters shall have an MTBF of at least 50,000 hours when predicted in accordance with MIL-HDBK-217 at 50°C.

3.7.2.11.4.2 Suppressed sideband attenuation. When operating in any available sideband mode, the suppressed sideband(s) shall be attenuated at least 60 dB below the level of the unsuppressed sideband(s).

3.7.3 Electronic keyer performance characteristics.

3.7.3.1 Keying speeds. The electronic keyer shall be capable of operation with CW keying up to 200 baud, FSK (TTY) up to 300 baud, and analog inputs (FAX) at speeds up to 400 baud.

3.7.3.2 Electronic keyer output frequency accuracy and stability. Over any 24-hour period, the accuracy and stability of all electronic keyer output tones shall be within ±1 Hz or ±1 percent, whichever provides greater accuracy and stability.

3.7.3.3 CW ON and OFF keying. The electronic keyer shall provide a 1000-Hz tone for CW ON and OFF keying.

3.7.3.4 Electronic keyer input signals. The electronic keyer shall accept the following types of signal input:

a. 20 mA neutral (external battery)

b. 60 mA neutral (external battery)

c. 50 V neutral (external battery)

d. 100 V neutral (external battery)

e. Dry relay contacts (internal battery)

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f. A FAX signal input which varies linearly between 1 V and 10 VDC and delivers a frequency shift tone which varies linearly from 400 Hz above to 400 Hz below the selectable rest frequency

g. Low level (±6 V) polar signals with input sensitivity of 0.5 V, positive mark, external battery, with no internal ground

3.7.3.5 Electronic keyer input impedances. The electronic keyer shall have the following input impedance for input signals specified in the corresponding subparagraphs of 3.7.3.4:

a. 150 ohms or less

b. 150 ohms or less

c. 47,000 ohms

d. 100,000 ohms

e. 47,000 ohms

f. 47,000 ohms minimum

g. 100,000 ohms or greater

3.7.4 Driver amplifier performance characteristics. The driver amplifier shall amplify linearly any emission presented to its input from the exciter within the limits of a 25-kHz maximum bandwidth. The power output delivered to a 50-ohm unbalanced nonreactive load over the frequency range of 2.0000 MHz through 29.9999 MHz shall be at least that specified in TABLE VI when driven by the input power specified for each mode of operation.

3.7.4.1 Driver amplifier avg noise power output rating. The minimum noise power output level of the driver amplifier shall be 200 W when three of the four nominal 3-kHz channels are modulated at the appropriate power level. Under this loading condition the intermodulation levels of 3.7.5.3.1 shall be met.

3.7.5 PA performance characteristics. The PA shall amplify linearly any emission presented to its input from the driver amplifier within the limits of a 25-kHz maximum bandwidth. The power output delivered to a 50-ohm unbalanced nonreactive load over the frequency range of 2.0000 MHz through 29.9999 MHz shall be at least as specified in TABLE VI when driven by the input power specified for each mode of operation.

3.7.5.1 Avg noise power output rating. The minimum avg noise power output level of the transmitter shall be 2 kW when three of the four nominal 3-kHz channels are modulated at the appropriate power level. Under this loading condition the intermodulation levels of 3.7.5.3.1 shall be met.

3.7.5.2 Out-of-band spurious emission. The power level of all out-of-band spurious emissions below the level of the unsuppressed level of the carrier frequency, including parasitic oscillations and intermodulation products outside of the normally transmitted bandwidth of 12.6 kHz, shall be as follows: any emission appearing on any frequency removed from the carrier frequency by at least 50 percent but not more than 100 percent of the maximum authorized bandwidth shall be attenuated not less than 43 dB; any spurious emission appearing on any frequency removed from the carrier frequency by at least 100 percent but not more than 300 percent of the maximum authorized bandwidth shall be attenuated not less than 60 dB. Any spurious or harmonic emission appearing on any frequency removed from the carrier frequency by at least 300 percent of the maximum authorized bandwidth shall be attenuated not less than 80 dB.

3.7.5.3 In-band intermodulation distortion. All in-band intermodulation products shall be measured from the audio inputs to the RF output of the transmitter. Requirements specified herein apply to any operating frequency from 2.0000 MHz to 29.9999 MHz.

3.7.5.3.1 In-band intermodulation distortion when white-noise loaded. When channels B_2, B_1, and A_2 (see FIGURE 1) are loaded with white-noise signals which produce 2kW avg power into a 50-ohm load connected to the RF output point of the transmitter, the avg power level measured in the nominal 3-kHz spectrum of channel A_1 shall be at least 43 dB below the 2kW avg power output of the transmitter. When the avg power output level of the transmitter is to be

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measured using a frequency domain indicating device in conjunction with a calorimetric measuring set or equivalent, a 4.77 dB factor derived from the formula:

$$dB = 10 \log \frac{P_1 + P_2 + P_4}{P_1}$$

may be added to the avg power difference between channel B_1 or any other loaded channel, and the avg power level P_3 appearing in channel A_1 to account for power integration in the 50-ohm termination resulting from equal power distribution in the three loaded channels. P_1, P_2, and P_4 are defined as: P_1 equals the avg power level appearing in the nominal 3-kHz spectrum of channel B_2 at the transmitter RF output point. The relationship between the total white-noise loaded power output of the transmitter (2kW avg) and the avg power appearing in the RF spectrum of channel B_2 or any other loaded channel (B_1 or A_2) is:

$$P_1 = \frac{2000}{3} = 666.66 \text{ W avg}$$

$$P_2 = P_1 \text{ and } P_4 = P_2$$

P_3 is the avg power appearing in the nominal 3-kHz spectrum of channel A_1 (the unloaded channel) at the transmitter RF output point when channels B_2, B_1, and A_2 are loaded with equal white noise signals such that 2 kW avg power output will result (see FIGURE 1).

3.7.5.3.2 Two-tone performance. When operated in the A3J mode at 10 kW PEP (5 kW avg power) with two tone signals at 1000 Hz and 1625 Hz applied to channel A_1, (USB) or channel B_1 (LSB) in accordance with the requirements specified in TABLE VI for full rated PEP (two-tone) output, all intermodulation distortion products shall be at least 35 dB below either tone appearing at the output point of the transmitter, including band-edge performance.

3.7.5.4 PA interference. Procurements of equipments and subsystems paragraph of MIL-STD-461 having met other EMI requirements, is applicable to this specification. The radiated and conducted interference produced by the PA shall not exceed the limits specified in MIL-STD-461. Measurement of the PA shall be made in accordance with MIL-STD-449.

3.7.5.5 In-band spurious emission and hum products. In-band spurious emission, including hum products, shall be at least 35 dB below the level produced by single tone modulation at a 5-kW power level. When any nominal 3-kHz channel is modulated with white-noise signals, the in-band spurious emission and hum products shall be at least 43 dB below the 2 kW avg power produced at the output of the transmitter.

3.7.5.6 Output impedance. The active output impedance of the transmitter shall be designed to operate into 50-ohms nominal. No degradation of specified performance shall occur when the external load impedance produces a VSWR of 2:1, or less, at the transmitter output fitting. No damage shall occur to any component of the transmitter when the external VSWR approaches the limit of 4:1. The transmitter shall be fitted with a standard 50-ohm, 4.1 cm (1.625 in.) flange fitting specified in EIA RS 225-59 at the RF output point.

3.7.5.7 Information bandwidth. With an RF input signal of constant level driving the PA at any frequency between 2.0000 MHz to 29.9999 MHz, with the automatic tuning system disabled, and with the PA terminated into a 50-ohm resistive load, it shall be possible to change the driving frequency by ±12.5 kHz without causing more than 1 dB variation in the PA output power level.

3.7.5.8 Regeneration. There shall be no unintended oscillation or instability under any condition of the driver amplifier and the PA when employing tubes with the maximum transconductance specified for the type tubes used and with a supply voltage up to 10 percent higher than the nominal primary voltage.

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3.7.5.9 Intermodulation, spurious emissions, and hum. Intermodulation products and spurious emissions produced in the PA shall be limited to levels consistent with the overall requirements specified in 3.7.5.2 for spurious emissions and in 3.7.5.3 for intermodulation distortion. In-band hum products shall be at least 43 dB below a 10-kW single-tone output reference level.

3.7.5.10 Harmonic output. Output of the second order harmonic shall be at least 80 dB below the power level of the fundamental signal at all power levels, including full rated output. Higher order harmonic products shall be more than 80 dB below the power level of the fundamental signal at all power levels including full rated output.

3.7.5.11 Keying transients. Keying transients originating in the transmitter shall be reduced to levels not exceeding those specified in 3.7.5.2 and 3.7.5.3.

3.7.5.12 Efficiency. The overall efficiency of the PA shall be no less than 25 percent determined from the 50 Hz or 60 Hz power input with the transmitter loaded with a two-tone signal to 10 kW PEP output.

3.7.5.13 PPC time constant. The PPC time constant shall be between 1 and 2 ms attack time with critical damping and between 60 ms and 100 ms decay time.

3.7.6 Remote control system interface performance characteristics.

3.7.6.1 LCU and exciter interface signal levels.

3.7.6.1.1 LCU to exciter. The signal level for the unselected state (logical 0) shall be from 0 V to + 0.5 V with a current sinking capability of 1.5 mA. The signal level for the selected state (logical 1) shall be from +1.5 V to +5.0 V at a maximum current of 100 microamperes (μA). The exciter control input circuitry shall provide a current sink for the LCU unit logic control circuitry. The 5.0 VDC for the interface circuitry contained in the LCU shall be provided by the exciter to minimize ground currents between units. A current of 10 mA shall not be exceeded.

3.7.6.1.2 Exciter to LCU. Unselected (logical 0) state equals closed circuit to LCU common line. When the status of the exciter is changed by the remote control system, monitoring information may provide invalid data for a period of up to 1.5 s.

3.8 Environmental conditions. The transmitter shall conform to the performance requirements of this specification under the following environmental conditions:

a. Operating:

1. Ambient temperature: Range 4 of MIL-E-16400, at sea level

2. Humidity: 0 percent to 95 percent RH

3. Elevation: Sea level to 3000 m (approximately 10,000 ft), 20°C at 50 percent RH

b. Nonoperating:

1. Ambient temperature: Range 4 of MIL-E-16400

2. Elevation: Sea level to 15,000 m (approximately 50,000 ft)

3.8.1 Dust. The transmitter shall operate without performance degradations or malfunctions when exposed to dust encountered in closed structures, including transportable vans.

3.8.2 Shock. The transmitter shall withstand shock encountered in handling and transportation without physical damage or performance degradations.

3.9 Dimensions. The transmitter shall be delivered in two cabinets with the following overall dimensions:

Height	Width	Depth
199 cm (78.313 in.)	Exciter-driver cabinet, 54 cm (21.25 in.)	101.6 cm (40 in.)
199 cm (78.313 in.)	Final power amplifier cabinet, 81.9 cm (32.25 in.)	101.6 cm (40 in.)

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3.10 Finish. Exterior cabinet surfaces shall be finished with enamel conforming to Type III, Class 2, of MIL-E-15090.

3.11 Identification plates. All major components of the transmitter shall have serial numbers and nomenclature. Identification plates and markings shall conform to MIL-E-21981.

3.12 Workmanship.

3.12.1 General Workmanship. The workmanship and soldering requirements shall be in accordance with MIL-E-16400.

3.12.2 Workmanship screen. All transmitters shall withstand a defect detection vibration screen of random type vibration at $0.04g^2$/Hz ±3 dB from 80 Hz to 350 Hz and temperature cycling as specified in 4.5.4. The final power amplifier is exempt from this requirement.

3.13 Support maintenance test procedure verification. When specified in the contract or order (see 6.2.1), the contractor shall perform a support maintenance test procedure verification as specified in 4.8.

3.13.1 Support maintenance. The support maintenance test procedure verification shall be performed by a technician selected by the Government from contractor or Government personnel.

3.13.2 Test equipment. Test equipment used to verify the support maintenance test procedures shall be selected from MIL-STD-1364. Test equipment not specified in MIL-STD-1364 may be used in the support maintenance test procedure verification only if the contractor submits a written request to the procuring activity and approval is granted.

3.13.3 Verification. The support maintenance test procedure verification shall be performed on a first article sample.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to a) Surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Quality conformance inspection:

1. In-process inspection (see 4.4.1)

2. Production inspection (see 4.4.2)

3. Periodic inspection (see 4.4.3)

c. Inspection of preparation for delivery (see 4.9)

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4.3 First article inspection. Unless otherwise specified (see 6.2.1), two units shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE VIII.

TABLE VIII. Examinations and tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				In-process inspection	Production inspection	Periodic inspection
Transmitter:						
Parts, materials, and processes	3.2.1	4.5.1	X			
Exciter-driver cabinet equipment arrangement	3.3.1.1	4.5.7	X			
Primary power	3.3.1.2	4.5.8	X			
Primary power cable entry	3.3.1.2.1	4.5.7	X			
Primary power connections	3.3.1.2.2	4.5.7	X			
Autotransformer	3.3.1.2.4, 3.3.1.2.4.1	4.5.7	X			
AM-6047B/FRT-84(V) primary power	3.3.1.2.5	4.5.7	X			
PP-6067A/FRT primary power	3.3.1.2.6	4.5.7	X			
Exciter, KY-655A/FRT, and LCU primary power	3.3.1.2.7, 3.3.1.2.7.1	4.5.7	X			
Circuit breakers	3.3.1.2.8	4.5.7	X			
Leakage current	3.1	4.5.5	X			
Accessibility	3.3.1.3	4.5.7	X			
Use of electron tubes and solid-state devices	3.3.1.4	4.5.7	X			
Capacitors	3.3.1.5 through 3.3.1.5.4	4.5.7	X			
Type BNC and Type N RF coaxial connectors	3.3.1.6	4.5.7	X			
Transformers	3.3.1.7	4.5.7	X			
PP-6067A/FRT and AM-6047B/FRT-84(V) screen rectifiers	3.3.1.8	4.5.7	X			
Insulating compounds	3.3.1.9	4.5.7	X			
Resistors	3.3.1.10	4.5.7	X			
Encapsulation	3.3.1.11	4.5.7	X			
Semiconductors	3.3.1.12	4.5.7	X			
Unacceptable materials	3.3.1.14	4.5.7	X			
Electrical contacts	3.3.1.15	4.5.7	X			
Relays	3.3.1.16	4.5.7	X			
Thermal design	3.3.1.17 through 3.3.1.17.2	4.5.9	X			
Acoustic noise	3.3.1.18	4.5.8	X			
Safety to personnel	3.3.1.19	4.5.1	X			
Interlocks	3.3.1.19.1	4.5.7	X			
Interlock circuit extension	3.3.1.19.1.1	4.5.8	X			
Capacitor discharge	3.3.1.19.2	4.5.7	X			

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TABLE VIII. Examinations and tests (Cont).

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				In-process inspection	Production inspection	Periodic inspection
Final power amplifier cabinet shorting stick	3.3.1.19.3	4.5.8	X			
AM-6046B/FRT shorting stick	3.3.1.19.4	4.5.8	X			
Overcurrent device location	3.3.1.19.5	4.5.7	X			
Protection:						
Protection from operator error	3.3.1.20.1	4.5.2	X			
External circuit fault protection	3.3.1.20.2	4.5.1	X			
Internal transient voltage protection	3.3.1.20.4	4.5.8	X			
Power supply protection	3.3.1.20.5	4.5.8	X			
PCB protection	3.3.1.20.6	4.5.7	X			
Test points	3.3.1.21	4.5.8	X			
Test cords and module extenders	3.3.1.22	4.5.7	X			
Meters and indicators	3.3.1.25, 3.3.1.25.1	4.5.7	X			
Controls	3.3.1.26, 3.3.1.26.1	4.5.7	X			
Wiring connections	3.3.1.27	4.5.7	X			
Construction:						
Modular construction	3.5.1	4.5.7	X			
Cabinet construction	3.5.2 through 3.5.2.5	4.5.7	X			
Degree of enclosure	3.5.3	4.5.2	X			
Mountings	3.5.4 through 3.5.4.7	4.5.8	X			
Environmental conditions	3.8	4.5.1	X			
Dust	3.8.1	4.5.10	X			
Shock	3.8.2	4.5.11	X			
Identification plates	3.11	4.5.7	X			
Weight marking	3.3.1.19.6	4.5.7	X		X	
VSWR protection	3.3.1.20.3	4.5.8	X		X	
Test jacks	3.3.1.21.1	4.5.8	X		X	
Interchangeability	3.3.8 through 3.3.8.5	4.5.8	X		X	
Frequency range	3.7.1.1, 3.7.1.1.1	4.5.8	X		X	
Operating modes	3.7.1.2	4.5.7	X		X	
Power output(s) for specified input(s)	3.7.1.3	4.5.8	X		X	
In-band electrical noise	3.7.1.5 through 3.7.1.5.2	4.5.8	X		X	
Envelope delay	3.7.1.6	4.5.8	X		X	
Dimensions	3.9	4.5.7	X		X	
Exciter:						
Exciter design requirements	3.3.2	4.5.7	X			
Exciter configuration	3.3.2.1	4.5.7	X			
Frequency selection	3.3.2.3	4.5.7	X			

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TABLE VIII. Examinations and tests (Cont).

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				In-process inspection	Production inspection	Periodic inspection
Synthesizer module	3.3.2.4 through 3.3.2.4.2	4.5.7	X			
TGC module	3.3.2.5	4.5.7	X			
Down-converter module	3.3.2.6	4.5.7	X			
Output amplifier module	3.3.2.7	4.5.8	X			
Exciter power supply	3.3.2.8	4.5.7	X			
Internal frequency standard	3.3.2.9.1	4.5.8	X			
Rear panel connector	3.3.2.9.1.1	4.5.7	X			
Front panel connector	3.3.2.9.1.2	4.5.7	X			
Internal frequency standard oven	3.3.2.9.1.3	4.5.7	X			
VFG hold time adjustment	3.3.2.10.2	4.5.8	X			
External PTT service	3.3.2.11	4.5.7	X			
Dry contact keying	3.3.2.13	4.5.7	X			
Transmitter status read-back function	3.3.2.14	4.5.7	X			
Front panel bulb voltage	3.3.2.15	4.5.7	X			
Exciter and KY-655A/FRT interface	3.3.2.16	4.5.7	X			
Exciter standby condition	3.3.2.17	4.5.7	X			
Two ISB simplex	3.3.2.18	4.5.7	X			
Controls-indicators: Frequency controls	3.3.2.19.1, 3.3.2.19.1.1	4.5.7	X			
Operating mode selection	3.3.2.19.2	4.5.7	X			
Amplifier OFF	3.3.2.19.5	4.5.7	X			
Standby	3.3.2.19.6	4.5.7	X			
Operate	3.3.2.19.7	4.5.7	X			
Tune	3.3.2.19.8	4.5.7	X			
Ready	3.3.2.19.9	4.5.7	X			
Local and remote operation selector	3.3.2.19.10	4.5.7	X			
Internal frequency standard calibration	3.3.2.19.11, 3.3.2.19.11.1	4.5.2	X			
Input level controls	3.3.2.19.12	4.5.8	X			
Exciter failure indicator	3.3.2.19.13	4.5.7	X			
Transmitter failure indicator	3.3.2.19.14	4.5.7	X			
Exciter circuit test meter	3.3.2.19.16, 3.3.2.19.16.1	4.5.7	X			
Exciter auxiliary control panel	3.3.2.19.17 through 3.3.2.19.17.3	4.5.7	X			
Signal inputs	3.7.2.3	4.5.7	X			
Audio impedance	3.7.2.3.1.2, 3.7.2.3.1.2.1	4.5.8	X			
Exciter output impedance	3.7.2.8	4.5.8	X			

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TABLE VIII. Examinations and tests (Cont).

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				In-process inspection	Production inspection	Periodic inspection
Four channel ISB performance	3.7.2.11.1	4.5.7	X			
Output channel inversion	3.7.2.11.1.1	4.5.7	X			
Channel frequency stability	3.7.2.11.1.2	4.5.2	X			
Inversion carriers and spurious suppression	3.7.2.11.1.3	4.5.7	X			
Sideband filter characteristics	3.7.2.11.4.1	4.5.7	X			
Exciter primary power	3.3.2.2	4.5.7	X		X	
External frequency standard	3.3.2.9.2 through 3.3.2.9.2.3	4.5.8	X		X	
VFG	3.3.2.10	4.5.8	X		X	
VFG keying line	3.3.2.10.1	4.5.7	X		X	
Automatic level control	3.3.2.12	4.5.8	X		X	
Sideband selector control	3.3.2.19.2.1	4.5.7	X		X	
Class of emission control	3.3.2.19.2.2	4.5.7	X		X	
Simplex 2 ISB selector	3.3.2.19.2.3	4.5.7	X		X	
Individual subchannel gain	3.3.2.19.3	4.5.8	X		X	
Exciter power output control	3.3.2.19.4	4.5.8	X		X	
VU input meter	3.3.2.19.15	4.5.7	X		X	
Frequency stability	3.7.2.1	4.5.8	X		X	
Aging	3.7.2.1.1.1.1	4.5.8	X			
Modulating signals (audio)	3.7.2.3.1 through 3.7.2.3.1.2.1	4.5.8	X		X	
Carrier suppression	3.7.2.4	4.5.8	X		X	
Carrier compression	3.7.2.5	4.5.8	X		X	
Carrier change under modulation	3.7.2.6	4.5.8	X		X	
Equal sideband power	3.7.2.7	4.5.8	X		X	
Standby power	3.7.2.9	4.5.8	X		X	
Transfer time and controls	3.7.2.10	4.5.8	X		X	
In-band response	3.7.2.11.2	4.5.8	X		X	
Input and output references	3.7.2.11.3, 3.7.2.11.3.1	4.5.8	X		X	
Out-of-band response	3.7.2.11.4	4.5.8	X		X	
Suppressed sideband attenuation	3.7.2.11.4.2	4.5.8	X		X	
Electronic keyer: KY-655A/FRT design requirements	3.3.3	4.5.7	X			
Electronic keyer primary power	3.3.3.1	4.5.7	X			
CW mode	3.3.3.3	4.5.7	X			
DC isolation	3.3.3.4	4.5.7	X			
Electronic keyer and exciter interface	3.3.3.5, 3.3.3.5.1	4.5.7	X			

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TABLE VIII. Examinations and tests (Cont).

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				In-process inspection	Production inspection	Periodic inspection
Front panel controls	3.3.3.6.1 through 3.3.3.6.1.3	4.5.7	X			
Electronic keyer internal contorls	3.3.3.6.2 through 3.3.3.6.2.3	4.5.7	X			
CW ON-OFF keying	3.7.3.3	4.5.7	X			
Electronic keyer input impedances	3.7.3.5	4.5.8	X			
CW, frequency shift, and analog keying	3.3.3.2	4.5.7	X		X	
Keying speeds	3.7.3.1	4.5.8	X		X	
Electronic keyer output frequency accuracy and stability	3.7.3.2	4.5.8	X		X	
Electronic keyer input signals	3.7.3.4	4.5.8	X		X	
Driver amplifier:						
AM-6046B/FRT design requirements	3.3.4	4.5.7	X			
Driver amplifier primary power	3.3.4.1	4.5.7	X			
Screen power supply protection	3.3.4.2	4.5.7	X			
Filament low voltage transformer circuit protection	3.3.4.3	4.5.7	X			
Driver amplifier final assembly	3.3.4.4, 3.3.4.4.1	4.5.7	X			
Tuning method	3.3.4.5 through 3.3.4.5.1.2	4.5.7	X			
Driver amplifier input and output impedances	3.3.4.6	4.5.8	X			
Driver amplifier standby	3.3.4.7	4.5.7	X			
Output coupling capacitor 1A1C4	3.3.4.8	4.5.7	X			
Screen and plate supply rectifiers	3.3.4.9	4.5.7	X			
Driver amplifier controls, meters, and indicators	3.3.4.10 through 3.3.4.10.5	4.5.7	X			
Driver amplifier RF sampling	3.3.4.11 through 3.3.4.11.2	4.5.7	X			
Driver amplifier performance characteristics	3.7.4	4.5.8	X			
Interconnecting box front panel indicators	3.3.5	4.5.7	X			
PP-6067A/FRT:						
PP-6067A/FRT design requirements	3.3.6, 3.3.6.1	4.5.7	X			
Driver amplifier avg noise power output rating	3.7.4.1	4.5.8	X		X	

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TABLE VIII. Examinations and tests (Cont).

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				In-process inspection	Production inspection	Periodic inspection
Final power amplifier: AM-6047B/FRT-84(V) design requirements	3.3.7	4.5.7	X			
PA Primary power	3.3.7.1	4.5.7	X			
PA power supply	3.3.7.2 through 3.3.7.2.2	4.5.7	X			
PA input impedance	3.3.7.6	4.5.8	X			
Final PA assembly	3.3.7.8, 3.3.7.8.1	4.5.7	X			
RF sampling	3.3.7.9 through 3.3.7.9.2	4.5.8	X			
PA standby	3.3.7.10	4.5.7	X			
Tuning capacitor 9A1C8	3.3.7.11	4.5.7	X			
RF loading capacitors 9A1C4 and 9A1C6	3.3.7.12	4.5.7	X			
DC isolation capacitors	3.3.7.13	4.5.7	X			
Capacitor 9A1C10	3.3.7.14	4.5.7	X			
RF coupling capacitors	3.3.7.15	4.5.12	X			
Tank inductors and RF coils	3.3.7.16	4.5.7	X			
RF switches	3.3.7.17	4.5.7	X			
Plate supply rectifier	3.3.7.18	4.5.7	X			
PA meters	3.3.7.19 through 3.3.7.19.6	4.5.7	X			
Final power amplifier cabinet meter panel indicators	3.3.7.20.3	4.5.7	X			
Tune up control indicators	3.3.7.20.4	4.5.7	X			
PA circuit breaker panel	3.3.7.21	4.5.7	X			
Band repeater assembly	3.3.7.22	4.5.7	X			
PA performance characteristics	3.7.5	4.5.8	X			
Output impedance	3.7.5.6	4.5.8	X			
Keying transients	3.7.5.11	4.5.8	X			
Efficiency	3.7.5.12	4.5.8	X			
PPC time constant	3.7.5.13	4.5.8	X			
Frequency range and tuning	3.3.7.3, 3.3.7.3.1	4.5.8	X		X	
TGC circuit	3.3.7.4	4.5.8	X		X	
PPC	3.3.7.5	4.5.8	X		X	
VSWR output monitor	3.3.7.7	4.5.8	X		X	
Manual tuning controls	3.3.7.20.1	4.5.7	X		X	
Tune detector assembly (9A1A3) controls	3.3.7.20.2	4.5.7	X		X	
Avg noise power output rating	3.7.5.1	4.5.8	X		X	
Out-of-band spurious emission	3.7.5.2	4.5.8	X		X	

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TABLE VIII. Examinations and tests (Cont).

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				In-process inspection	Production inspection	Periodic inspection
In-band intermodulation distortion	3.7.5.3 through 3.7.5.3.2	4.5.8	X		X	
In-band spurious emission and hum products	3.7.5.5	4.5.8	X		X	
Information bandwidth	3.7.5.7	4.5.8	X		X	
Regeneration	3.7.5.8	4.5.8	X		X	
Intermodulation, spurious emissions, and hum	3.7.5.9	4.5.8	X		X	
Harmonic output	3.7.5.10	4.5.8	X		X	
Remote control system	3.3.9 through 3.3.9.3.1	4.5.7	X			
LCU and exciter interface signal levels	3.7.6.1 through 3.7.6.1.2	4.5.7	X			
Reliability qualification test	3.4.1	4.6	X			
Maintainability demonstration	3.6.1	4.7	X			
Support maintenance test procedure verification	3.13 through 3.13.3	4.8	X			
Cables	3.3.1.13	4.5.6	X	X		
Retractable cable harness	3.5.5	4.5.7	X	X		
Finish	3.10	4.5.7	X	X		
General workmanship	3.12.1	4.5.3	X	X		
Workmanship screen: Random vibration	3.12.2	4.5.4.1	X	X		
Temperature cycling	3.12.2	4.5.4.2	X	X		
Electromagnetic field	3.7.1.4	4.5.8	X			X
Interference	3.7.2.2	4.5.8	X			X
PA interference	3.7.5.4	4.5.8	X			X

FAS tests shall be conducted with the Government-furnished AN/FSQ-98(V) or AN/FSQ-98A(V) installed and functioning as designed, if Government-loaned property is provided. The FAS shall be configured for 4 ISB operation, including the necessary side carrier generator, modulators and filters.

4.4 Quality conformance inspection.

4.4.1 In-process inspection. Each transmitter shall be subjected to the in-process inspection specified in TABLE VIII.

4.4.2 Production inspection. Production inspection shall be as specified in TABLE VIII. Production inspection shall be performed on a sampling basis in accordance with the inspection procedures of MIL-STD-105, using the special inspection level S-3 for normal, tightened, and reduced inspection. Production inspection shall be performed on transmitters that have completed the in-process inspections. Lots shall be physically accumulated and identified prior to random selection of the sample. The Government shall make the random sample selection. Sample transmitters shall satisfactorily conform to the production inspection requirements prior to releasing the lot for shipment. The acceptable quality level (AQL) shall be 6.5 percent for each attribute.

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4.4.2.1 Rejected lots. If an inspection lot is rejected, the contractor may rework the rejected lot to correct the defective transmitters and reinspect the lot, using tightened inspection. As a minimum, the rework by the contractor shall consist of inspecting each transmitter in the rejected lot for the failed attribute prior to reinspection using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.3 Periodic inspection. Periodic inspection shall be as specified in TABLE VIII and shall be performed on units which have been subjected to, and have passed, production inspection.

4.4.3.1 Sampling for periodic inspection. One transmitter shall be selected from each successive 50 transmitters, or fraction thereof, produced. The first sample transmitter shall be selected from the first month's production.

4.4.3.2 Nonconforming periodic sample units. If a sample unit fails the inspection specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the Government quality assurance representative (QAR) the results thereof and details of the corrective action taken to correct units of product which were manufactured under the same conditions, with the same materials, processes, and so forth. If the QAR does not consider that the corrective action will enable the product to conform to specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

4.4.3.3 Reinspection of conforming periodic sample units. Unless otherwise specified (see 6.2.1), sample units which have been subjected to, and have passed periodic tests, may be accepted on the contract provided they are resubjected to, and pass, production inspection specified in 4.4.2 after any necessary rework.

4.5 Test methods.

4.5.1 MIL-E-16400 tests. When 4.5.1 is specified in TABLE VIII, the test method specified in MIL-E-16400 shall be applied.

4.5.2 Test by analysis. When 4.5.2 is specified in TABLE VIII, the test may be performed by analysis, or calculation, or both.

4.5.3 General workmanship. The transmitter, including subassemblies and assemblies, shall be examined for workmanship and soldering during the fabrication and assembly process for conformance to the requirement of 3.12.1. Each solder connection and its associated wiring or leads shall be visually examined.

4.5.4 Workmanship screen.

4.5.4.1 Vibration. Prior to conducting temperature cycling, vibration shall be performed on each transmitter. The vibration may be performed at the module, drawer, or end item level. All the hardware, including cables and connectors, shall be exposed to vibration.

4.5.4.1.1 Type of vibration. The vibration shall be random, or subject to procuring activty approval, pseudo-random or complex waveform vibration, for an accumulated time of 10 minutes in the axis deemed most susceptible to vibration excitation. All test items shall be hard-mounted (without shock isolators) and subjected to the vibration conditions of FIGURE 2. The control accelerometer shall be located next to one of the mounting points of the transmitter under test. Transmitters having a bandwidth no greater than 10 Hz for vibration frequencies up to 500 Hz and 100 Hz for vibration frequencies above 500 Hz shall be used for the control and analysis of the acceleration spectral density (ASD).

4.5.4.1.2 Vibration test conditions. The test transmitter shall be energized during vibration and appropriate input signals applied to observe any abnormal conditions of the output functional characteristics. All failures occurring during the test shall be corrected and the test resumed.

4.5.4.2 Temperature cycling. Each transmitter shall be subjected to 10 cycles of the temperature curve shown in FIGURE 3. The temperature rate of change shall be not less than 5°C per minute. Transmitter power shall be turned on and off at the indicated times. When practical, the equipment drawers, panels, and enclosures shall be opened or removed for maximum exposure to the changing temperature. When performance measurements are called for, a functional operating test shall be performed. The dwell time shall be 80 percent of the time required for the largest electrical or electronic part to become temperature stabilized. Temperature stabilization shall be as defined in MIL-STD-810 for the nonoperating condition. All failures occurring during test shall be corrected and the test resumed.

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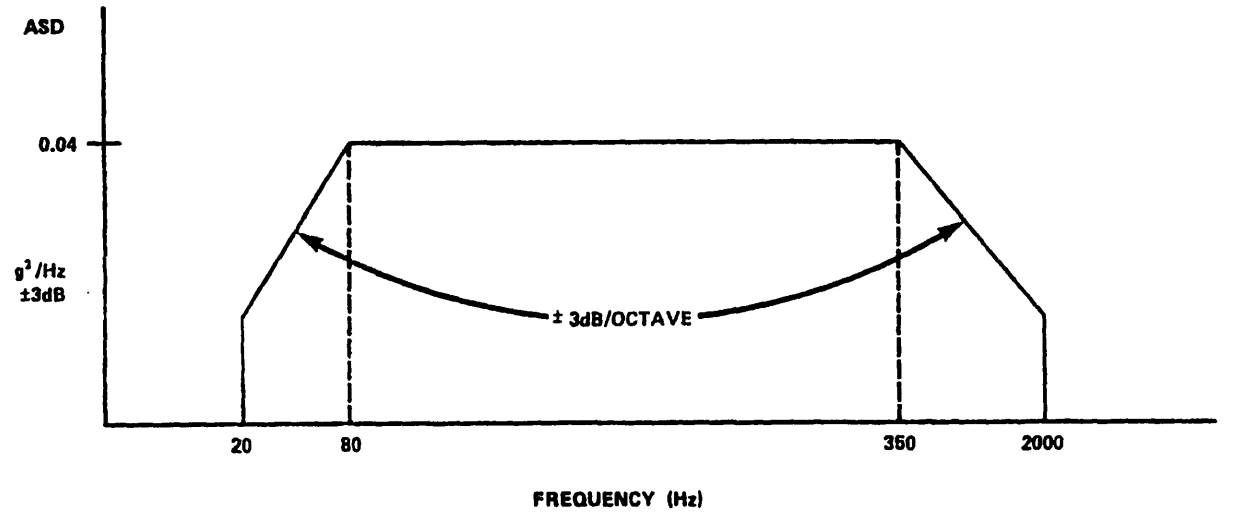


FIGURE 2. Random vibration curve.

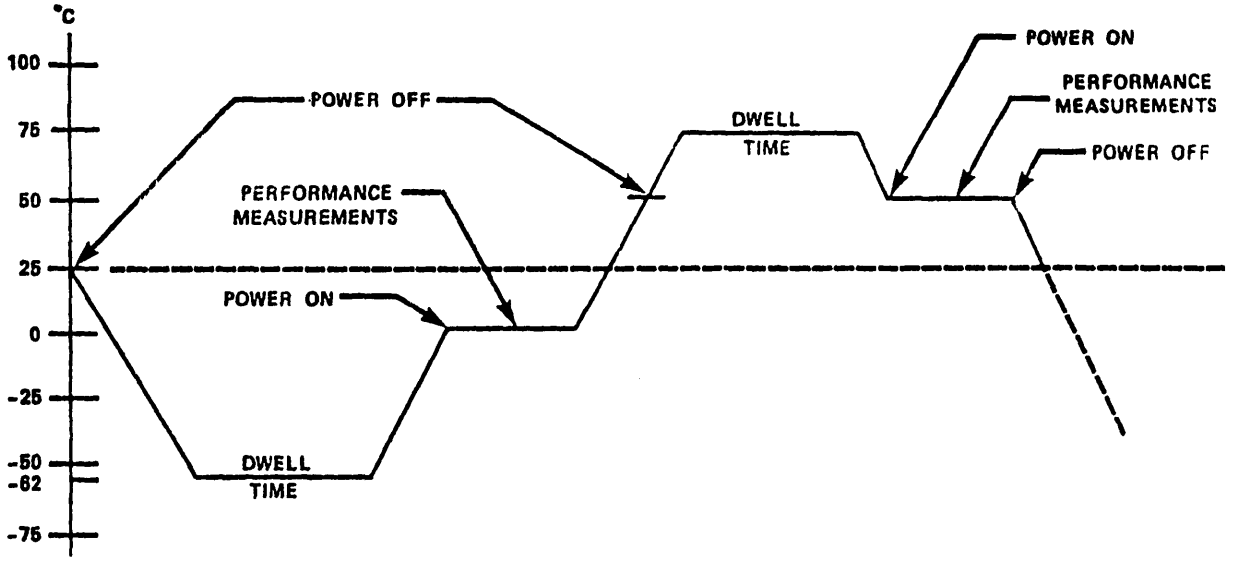


FIGURE 3. Temperature curve (one cycle).

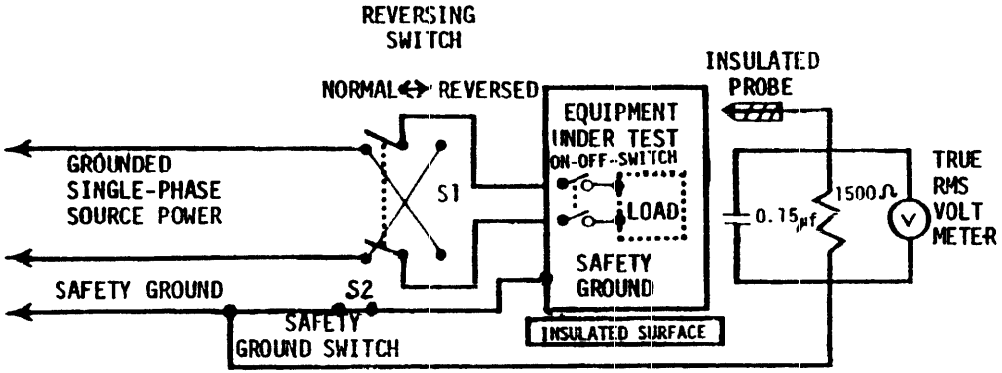
4.5.5 Leakage current test.

WARNING

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE TRANSMITTER DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES. THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

4.5.5.1 Transmitter connections. Each transmitter directly connected to an external power source and units deriving power from the transmitter shall be placed on an insulated surface. All safety ground conductors between the transmitter and units deriving power from the transmitter shall be intact. The safety ground conductor between the transmitter under test and the source power shall be opened during the test. The transmitter shall be connected as shown in FIGURE 4 if it is connected to single-phase source power, and as shown in FIGURE 5 if connected to three-phase source power.

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WARNING

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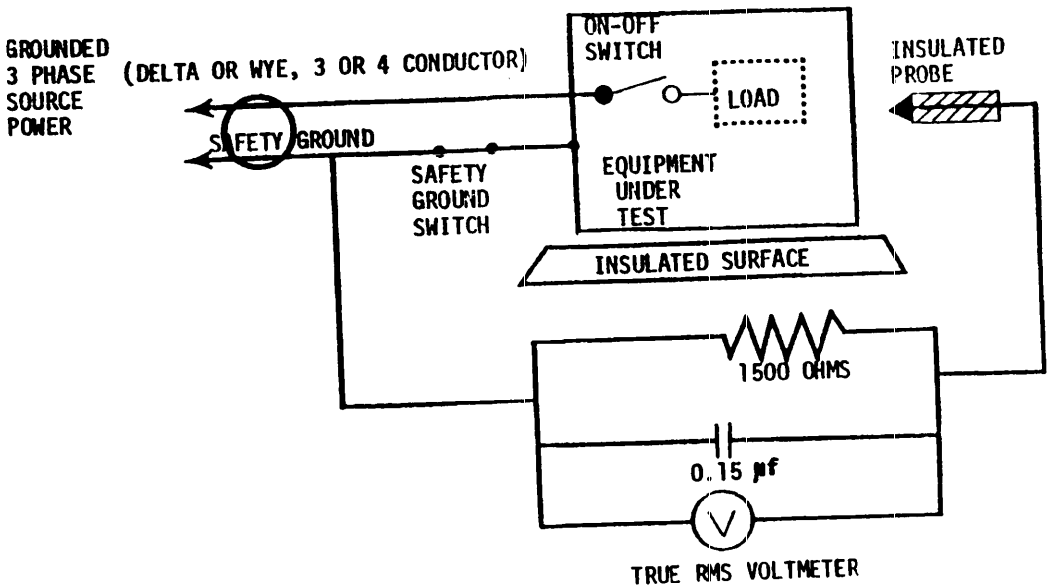
THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR FROM THE USE OF THIS DIAGRAM FOR LEAKAGE CURRENT MEASUREMENT.

GENERAL ORDER OF TEST:

1. SOURCE POWER OFF. CONNECT EQUIPMENT PER DIAGRAM.
2. ON-OFF SW OFF. S1 SW NORMAL. S2 SW CLOSED. CONNECT SOURCE POWER.
3. OBSERVE WARNING S2 SW OPEN. ON-OFF SW ON.
4. FOR EACH PROBE POINT, RECORD VOLTMETER READING (CASE, CONNECTORS, CONTROLS, SHAFTS).
5. ON-OFF SW OFF. REPEAT STEP 4.
6. S1 SW REVERSED. ON-OFF SW ON. REPEAT STEP 4
7. ON-OFF SW OFF. REPEAT STEP 4.
8. S2 SW CLOSED. S1 SW NORMAL.
9. REPEAT STEP 3 THRU 8 FOR EACH MODE OF OPERATION.
10. REMOVE SOURCE POWER. DISCONNECT EQUIPMENT. SIGN RECORD SHEET.

FIGURE 4. Single-phase test diagram for leakage current measurement.

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GENERAL ORDER OF TESTS:

1. SOURCE POWER OFF. CONNECT EQUIPMENT PER DIAGRAM.
2. ON-OFF SW OFF. SAFETY GROUND SWITCH CLOSED. CONNECT SOURCE POWER.
3. OBSERVE WARNING. SAFETY GROUND SWITCH OPEN. ON-OFF SW ON.
4. FOR EACH PROBE POINT, RECORD VOLTMETER READING (CASE, CONNECTORS, CONTROLS, SHAFTS).
5. ON-OFF SW OFF. REPEAT STEP 4.
6. SAFETY GROUND SWITCH CLOSED.
7. REPEAT STEP 3 THRU 6 FOR EACH MODE OF OPERATION
8. REMOVE SOURCE POWER. DISCONNECT EQUIPMENT. SIGN RECORD SHEET.

NOTE: 1. All three phases shall be connected during measurement.
2. The safety ground conductor shall not carry load current.

WARNING
DO NOT TOUCH EXPOSED METAL SURFACES
THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR FROM THE USE OF THIS DIAGRAM FOR LEAKAGE CURRENT MEASUREMENT.

FIGURE 5. Three-phase test diagram for leakage current measurement.

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4.5.5.2 Measurement. Leakage current shall be measured on the transmitter in its normal operating configuration. Transmitter controls in each operating mode shall be such that maximum power will be utilized during leakage current measurements. The leakage current shall be determined by the voltage-drop method. A true rms voltmeter shall be used. The voltage measured across the 1500-ohm resistor shall not exceed 7.5 V at the highest nominal power line voltage and the highest and lowest nominal power line frequencies for which the transmitter is designed. The overall measurement error shall not exceed five percent. The probe shall be used on all external conducting parts such as case, connector housings, recessed calibration or adjustment controls, and control shafts with knobs removed, and the voltage measured for every combination of switch positions available in FIGURES 4 and 5. The open safety ground conductor shall be reconnected immediately after the test is completed.

4.5.6 Cable tests. Cables to be supplied with the end item that are not captive to the transmitter shall, as an assembly, be subjected to a continuity and voltage test. The continuity test shall ensure that wires terminate on the correct connector pins. The voltage to be applied to each wire or cable for the voltage test shall be equal to the highest voltage rating, without exceeding the voltage rating for any constituent part, of the cable assembly.

4.5.7 Contractor validation. When 4.5.7 is specified in TABLE VIII, the contractor shall inspect the transmitter to determine technical compliance and adequacy for the specified characteristic or requirement. This validation shall include qualitative examination and testing whenever a function is specified.

4.5.8 Inspection by testing. When 4.5.8 is specified in TABLE VIII, the contractor shall perform quantitative testing.

4.5.9 Thermal survey testing. When specified in the contract or order (see 6.2.1), the contractor shall conduct a thermal survey. The survey shall be performed on those parts dissipating 1 percent, or more, of the total power dissipation of the unit in which the part is located and to ensure that heat sensitive parts are within the thermal stress identified in the reliability prediction report and the part derating requirement.

4.5.9.1 Survey conditions. The thermal survey shall be made under the following conditions:

a. Maximum operating temperature range (see 3.8.a.1)

b. Operating mode(s) which will cause the maximum steady-state power dissipation

c. Continuous transmitter operation for a time span that will achieve thermal stabilization

d. Cooling system operational, if other than natural methods

e. Transmitter cabinets closed

4.5.9.2 Proper techniques. Proper techniques such as thermocouples, infra-red photograpny, chemicals, or calibrated thermal sensitive materials that will accurately measure temperature shall be used in the survey. The method used to measure temperature shall not affect the accuracy of the measurement as a result of the heat sink effect or for any other reason.

4.5.10 Dust test. The transmitter shall be subjected to the dust test of MIL-STD-810, Method 510.1 (except that testing in the operating condition is excluded and the ambient temperature shall not exceed 50°C), without performance degradations or malfunctions to verify conformance with 3.8.1.

4.5.11 Shock test. The transmitter shall be subjected to the shock test of MIL-STD-810, Method 516.2, Procedure I, and the impact test listed in the table entitled Rough handling tests of MIL-P-116 without performance degradations or malfunctions, to verify conformance with 3.8.2.

4.5.12 RF coupling capacitor test. Testing of RF coupling capacitors under a peak stress voltage of 20,000 VDC shall be conducted continuously at 50°C for a period of not less than 15 minutes to verify conformance with 3.3.7.15.

4.6 Reliability qualification test. When specified in the contract or order (see 6.2.1), a reliability qualification test shall be performed in accordance with Test Plan XXC of MIL-STD-781 on the two FAS. Environmental conditions and operational duty cycles shall be as specified in 4.6.1 through 4.6.5.

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4.6.1 Environmental conditions for reliability testing. The environmental conditions shall be as follows:

a. Input voltage : Nominal ±10 percent

b. Vibration: None

c. Temperature: +50°C

d. Moisture: 75 percent RH

4.6.2 Performance tests. At least daily, performance measurements shall be made to verify the following performance characteristics:

a. Power out

b. Frequency and long term frequency stability

c. Modulation

d. Intermodulation

4.6.3 Test frequencies. The following test frequencies shall be used:

a. F_1: 2.0005 MHz

b. F_2: 4.9995 MHz

c. F_3: 10.0019 MHz

d. F_4: 29.9999 MHz

4.6.4 Test schedule. The test shall consist of three 8-hour schedules per day, the first being manned, where performance measurements shall be made. The other two schedules may be unmanned and pertinent data may be automatically recorded to the extent that instrumentation permits. These schedules shall be as follows:

a. Manned schedule:

1. Zero to 2 hours: Transmitter on
2. Two hours to 3 hours: Transmitter performance measurements
3. Three hours to 4 hours: Transmitter standby
4. Four hours to 6 hours: Transmitter on
5. Six hours to 7 hours: Transmitter measurements
6. Seven hours to 8 hours: Transmitter standby

b. Unmanned schedule:

1. Zero to 3 hours: Transmitter on
2. Three hours to 4 hours: Transmitter standby
3. Four hours to 7 hours: Transmitter on
4. Seven hours to 8 hours: Transmitter standby

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4.6.5 Transmitter schedules. The mode of operation and frequency of transmission for each transmitting ON period of the manned schedules shall occur in the following sequence:

Day	Frequency	Mode of transmission (into a 50-ohm dummy load)
1	F_1: First 4 hours of manned schedule	AM (A3E), 50 percent rated avg carrier 30 percent to 50 percent, 1000 Hz modulation; transmitting for 10 minutes out of each 15-minute period
	F_2: Remainder of the day	
2	F_3: First 4 hours of manned schedule	CW (A1), full rated avg power keyed at speed of 25 dot Hz; transmitting for 10 minutes out of each 15-minute period
	F_4: Remainder of the day	
3	F_1: First 4 hours of manned schedule	FSK (F1), full rated avg power keyed from space to mark at a 30-Hz rate (RYs); transmitting for 10 minutes out of each 15-minute period
	F_2: Remainder of the day	
4	F_3: First 4 hours of manned schedule	ISB (A3B), 100 percent of rated PEP, two-tone modulation per sideband; transmitting for 10 minutes out of each 15-minute period
	F_4: Remainder of the day	

The same sequence shall be reused on the next four days, except that the sequence of frequencies shall be reversed: F_4 and F_3 on the fifth day, F_2 and F_1 on the sixth day, F_4 and F_3 on the seventh day, and F_2 and F_1 on the eighth day. These schedules shall be repeated as necessary.

4.7 Maintainability demonstration. When specified in the contract or order (see 6.2.1), the contractor shall perform a maintainability demonstration at the organizational level. The demonstration shall be performed by qualified technicians, and shall be used to verify that the transmitter M_{ct} requirements are fulfilled.

4.7.1 Demonstration conditions. The transmitter maintainability requirements shall be demonstrated in accordance with MIL-STD-471 by replacement of subassemblies (modules, individual printed wiring boards (PWBs) or PCBs) and chassis-mounted electronic, electrical, electromechanical, and mechanical components or parts at the organizational level. Fifty candidate corrective maintainability faults shall be determined in accordance with Appendix A of MIL-STD-471. The following information shall be made available for each candidate task:

a. Designation of specific faulty part

b. Failure mode

c. Means of introducing fault (substitution of faulty part or simulation thereof)

The maintainability demonstration shall be conducted under the standard test conditions of MIL-E-16400. The contractor shall use only that test equipment, tools, jigs, fixtures, software programs, and technical documentation that are used in the manufacture and checkout of the transmitter as described in the maintainability demonstration plan.

4.7.2 Accept or reject criteria. The accept or reject criteria for the demonstration of the corrective maintenance times shall be as shown in TABLE IX.

TABLE IX. Acceptance criteria.

	M_{ct}	M_{maxct}
Acceptance level	8	0
Sample size	20	20

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Acceptance shall occur when the number of observed corrective maintenance task times which exceed the required value for each index (M_{ct}, M_{maxct}) is less than or equal to that shown in TABLE IX, corresponding to each index for the specified confidence level. The duration of each task shall be compared to the required value(s), and recorded as greater or lesser than each index value. The actual value obtained shall also be recorded. An acceptance decision can only be made when the values obtained for both M_{ct} and M_{maxct} have been accepted. The accept or reject criteria for preventive maintenance are based on the capability of performing each preventive maintenance task with no degradation in transmitter performance. Occurrence of transmitter performance degradation shall be cause for rejection of a preventive maintenance task.

4.7.3 Rejection. If the inspection results in a rejection, the contractor shall report to the QAR assigned to the plant, the results thereof and details of the corrective action to be taken. If the QAR does not consider that the corrective action will enable the product to conform to the specified requirements, the matter shall be referred to the contracting officer.

4.8 Verification of support maintenance test procedure. When specified in the contract or order (see 6.2.1), the contractor shall perform a verification of support maintenance test procedure conforming to the following:

a. A technician shall be given a complete set of boards and modules, all of which are completely unaligned.

b. The technician shall align each of the modules and boards and verify that each board or module is operating properly. These tests shall be done with the modules on the bench (modules or boards removed from the prime equipment).

c. The technician shall install the modules or boards in the prime equipment and align the prime equipment as specified in the manual.

d. The technician shall operate the equipment as specified in the approved test procedure.

4.9 Inspection of preparation for delivery. Inspection shall be performed to ensure conformance with the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2.1).

6. NOTES

6.1 Intended use. The transmitter is a Navy general-purpose shore-based radio transmitter for point-to-point high frequency communications.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Thermal survey (see 3.3.1.17.2 and 4.5.9)

c. When SPETE and special tools are required (see 3.3.1.24)

d. Requirement for configuration of exciter for 4 ISB operation (see 3.3.2.1)

e. Interchangeability requirements, if other than specified in 3.3.8

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f. Support maintenance test procedure verification (see 3.13 and 4.8)

g. Number of first article samples to be submitted if other than specified in 4.3

h. Reliability qualification test (see 4.6)

i. Maintainability demonstration (see 4.7)

j. When reinspected periodic sample units may not be accepted (see 4.4.3.3)

k. Levels of preservation, packaging, packing, and marking required (see 5.1)

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Defense Acquisition Regulation (DAR), the data requirements identified herein will be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL) (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data specified herein will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data requirement	Applicable DID
3.2.1.3	Nonstandard Part Approval Request	DI-E-7028
3.3.1.20.4	Test Report, Internal Transient Voltage Survey	DI-T-23729
4.5.9	Thermal Survey Report	DI-R-7036
4.3, 4.4, and 4.5	Test Procedures	UDI-T-22710
4.6	Reliability Test and Demonstration Reports	DI-R-7034
4.6	Reliability Test Procedures	DI-R-7035
4.7	Plan, Maintainability Demonstration	DI-R-2129/UDI-R-20440
4.7	Maintainability Demonstration Report	DI-R-2130
4.8	Support Maintenance Test Procedure Verification Report	UDI-T-22727

(Copies of DIDs required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 First article. When a first article is required it shall be tested and approved under the appropriate provisions of 7-104.55 of the DAR. The first article should be a first production item. The first article should consist of one unit. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests, and approval of the first article.

6.4 Definitions. Definitions of terms used in this specification are given in 6.4.1 through 6.4.6.

6.4.1 In-band electrical noise. In-band electrical noise is defined as any signal present in the transmitter output, the frequency of which, is removed from the carrier frequency by no more than 6.3 kHz when the transmitter is conditioned for normal operation with the four audio inputs terminated resistively in their characteristic impedance of 600 $\pm j$ zero ohms and no input signal is applied.

6.4.2 In-rush current. In-rush current in the transmitter consists of the AC input current transients generated in the primary side of the high-voltage power transformer during the initial period of 106.38 ms following the application of AC line voltage to the power transformer.

6.4.3 Absolute frequency error. Absolute frequency error is the total deviation in Hz of the output carrier frequency (present in the output or fully suppressed) from that frequency indicated on the exciter frequency selection controls or their associated indicating devices.

6.4.4 Frequency calibration. Frequency calibration of the exciter is defined as a process for:

a. Determining the frequency stability

b. Determining the absolute frequency error

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Insulation resistance 100%

c. Reducing the absolute frequency error to less than 0.3 Hz at any frequency covered by this transmitter

6.4.5 Exciter standby condition. Exciter standby condition is defined as that condition wherein the maximum amount of power consuming circuitry in the exciter is deactivated while all circuitry which affects the stability and accuracy of the exciter output frequency remains in full operating condition.

6.4.6 Protected. An open enclosure in which all ventilated openings other than those for cooling air inlets and exhausts are limited in size and shape. The openings should be of such shape as to prevent the passage of a rod larger than 12.7 millimeters (mm) (0.50 in.) in diameter.

6.5 GFE and GFM. The contracting officer should arrange to furnish the GFE and GFM specified in 3.3.1, 3.3.2, 3.3.4, 3.3.7, and 3.5.5.

6.6 Special tools. Special tools are defined as those tools not listed in the National Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service (DCAS)).

6.7 Changes from previous issues. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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